

With regard to **electrical contractors** working on site, as far as is practical, the site management team must assure themselves of the following.

- The contractor has compiled an accurate assessment of the risks to their employees and other people who may be affected by the work.
- The people at risk are aware and know what to do to protect themselves.
- An objective decision has been taken over whether the job can be carried out with the circuit(s) live or dead.
- The work will be properly planned, which includes the contractor informing site management of the circuit(s) to be worked on, and the implications for others (for example, if they have to disable a fire detection system).
- The job will be adequately supervised, with permanent accompaniment where necessary.
- The contractors will implement a permit to work system, where justified by the risks. The person issuing this permit must be competent, and have the appropriate technical knowledge and experience.
- A written safe system of working (incorporating the method statement) will be produced, communicated to those doing the job, and followed.
- The contractor will provide briefings to their employees and to others, as necessary.
- The contractor has employed only competent employees.
- The contractor will provide the correct tools and test equipment.
- The contractor's employees will be provided with suitable and sufficient PPE.

Where it is necessary to work on **live equipment**, it will be necessary for the person doing the work to be accompanied by another person. It is essential that this accompanying person:

- can recognise the signs of danger
- is able to administer first aid for electric shock
- can otherwise give assistance in the event of an emergency
- knows how to switch off the supply
- can assist in restricting access to the area.



High-risk work should always be carried out in accordance with a permit to work system.



As an indicator of competence the National Inspection Council for Electrical Installation Contracting (NICEIC) provides assessment and certification services for contractors working across the building services sector.



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Temporary works

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Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



**Scan the QR code to complete a brief survey,
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Overview

The requirement to manage temporary works can be misunderstood and many believe it is only required on large projects or for complex work. However, temporary works are likely to form a vital part of any construction project.

The Health and Safety Executive (HSE) report *Preventing catastrophic events in construction* identified that insufficient consideration is given to the management of temporary works. This leaves projects open to risks, such as injuries or fatalities, the failure or collapse of both temporary and permanent works, damage to adjacent premises, and the consequent delays and increased costs.

Temporary works design, planning, execution and interfaces with plant and equipment must all be taken seriously.

This chapter aims to raise awareness of temporary works and the legal requirements to manage them.

4.1 Introduction

An important element of all construction projects relates to those items used to assist in the construction, but which do not usually form part of the final (permanent) works.

Temporary works are described in BS 5975:2019, clause 5.1.1.1 as:



providing an “engineered solution” that is used to support or protect either an existing structure or the permanent works during construction, or to support an item of plant or equipment.

Temporary works are wide ranging, but some examples are shown below.

- Site hoardings and signboards.
- Temporary access roadways and working platforms for plant.
- Temporary earthworks.
- Reinforcement.
- Excavations and excavation supports.
- Propping systems for removal of load bearing walls and façade retention.
- Scaffolding and other proprietary systems used for access or support.

Two other terms related to temporary works are given below.

Falsework, which is the temporary support given to structures until they are able to sustain their own loads.

Formwork, which is the temporary form/mould (usually steel or timber) used to contain concrete until it has attained sufficient strength to withstand its own weight.

Although, on completion, it is the permanent works that receive attention, without temporary works being properly thought out, planned and managed, the project will not be successful. Effective, safe and perhaps imaginatively designed temporary works solutions can bring significant benefits and improve projects in terms of safety, certainty, productivity, efficiency, quality, completion times and cost.

On the other hand, poorly planned, designed, managed, supervised or constructed temporary works leave projects open to risks such as injuries or fatalities, the failure or collapse of both temporary and permanent works, damage to adjacent premises, and the consequent delays, increased costs and possible criminal charges.

4.2 Important points

Temporary works are important for several reasons.

- They can form a significant part of the overall cost of the project.
- They are often safety-critical (for example, if they fail, there is a high probability that people will be injured or killed).
- Often they have important characteristics that do not apply to permanent works (*refer to 4.3*).
- They have the potential to cause delay and additional cost if not planned in adequate detail.

As for all aspects of construction projects it is important that, at all stages:

- all parties have adequate competence, resource and information to fulfil their role
- it is clear who is responsible for what
- all parties on site are aware of where their responsibilities start and finish.

This is particularly critical at interfaces, between temporary works and permanent works, between different elements of temporary works or where there are abnormal loads in the permanent works during construction.

e.g. Tower crane

Consider the following situation.

The permanent works foundation is to be used as a base for a tower crane. Some modifications of this foundation are required to accommodate the tower crane fixings.

Element	Designation	Comment
Foundation	Permanent works	Used as temporary works to found tower crane (with permanent works designer approval).
Foundation adaptation	Temporary works	Included to facilitate fixings; will become permanent works if it is not removed on completion (with permanent works designer approval).
Fixings into base	Temporary works	Co-ordinated with foundation layout and reinforcement details. Requires design check and co-ordination with permanent works designer.
Tower crane	Plant	Removed on completion.

The elements described above will involve several parties: permanent works designer, temporary works designer, contractor (foundation), supplier (fixings) and supplier (crane).

Unless the design and construction interfaces between these organisations and activities are adequately managed, and considered in advance of the work, delay or failure could result.

4.3 Characteristics of temporary works

Temporary works often have characteristics that are not present in permanent works or which present themselves differently.

You can view these in the table below.

Construction accuracy	Although some temporary works have specified/recommended accuracy (specifically in respect of how upright they should be), some do not. Experience shows that, generally, accuracy is less likely to be achieved with temporary works than with permanent works. However, accuracy can be important in order to achieve the assumed strength and robustness.
Misalignment of components	Assemblies made up of small components (such as proprietary falsework towers) may incorporate misalignments between sections.
Reuse	Unlike the permanent works, temporary works components are reused and hence the risk of damage, fatigue cracking or distortion should be recognised and managed.
Changes	Unauthorised changes (for example, in the layout or by the removal of components that obstruct trades) can seriously compromise the safety of temporary works (such as scaffolds).
Abuse	Temporary works can be prone to abuse in the form of overloading, impact and poor quality control (such as welding), unless adequately managed.
Utilisation	As temporary works are designed for a specific task, they are usually loaded to their full design load. Permanent works are rarely loaded to this extent.

Site management procedures should recognise these characteristics to ensure that none occur to an extent that would be detrimental to the effective use or safety of the temporary works being used.

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4.4 Responsibilities

Responsibilities for temporary works arise in two ways: via statutory law and through contract.

4.4.1 Statute

Statutory requirements are legally binding and they override any contractual stipulations. There is no differentiation between temporary works and permanent works under the law: all hazardous work must be managed safely.

The main requirements, which stem from the Construction (Design and Management) Regulations 2015 (CDM), are set out, in summary, in the following table.

Communication, co-operation and co-ordination	All parties must communicate, co-operate and co-ordinate to ensure that temporary works are safe, and will not endanger or cause unnecessary ill effects upon the health of those involved.
This requirement is common sense. It will not only help to avoid accidents and ill health but also is an essential ingredient for a successful project. Contract requirements need to set out how this can be achieved (such as site meetings and information flow mechanisms). Risk thrives at interfaces.	
Competency	It is obligatory on all parties to be satisfied that they have the training, knowledge, skills, resources and adequate supervision to carry out the tasks assigned to them, and for the client to be equally satisfied.
The employment of a sub-contractor or designer who is not capable of safe or efficient working, or lacks experience, will sooner or later create a liability. It is important to have control over any further sub-contracting by sub-contractors themselves.	
Timing and resources	All parties must be afforded sufficient time to plan, mobilise and execute the works. They should also be provided with adequate resources. This includes those involved in the design or execution of temporary works. Hence, good project planning is essential.
To facilitate early and effective planning, designers and those who will execute the work must be provided with pre-construction phase information to establish interdependencies, constraints, sequencing and resource requirements.	
Pre-construction information	Under CDM, the principal contractor or contractor (if only one) should have received pre-construction information from the client, the principal designer or the design team; some of this may be relevant to temporary works design or execution. If so, it is important that it is passed to those sub-contractors or designers who are involved. Ensuring the timely distribution of all necessary information is one of the duties of the principal designer. This duty is also imposed on the principal contractor.
One of the recurrent complaints from sub-contractors is that they do not receive such information.	
Construction phase plan	This is the document that records the health and safety rules and management arrangements relating to the construction phase of a project and includes any temporary works.
Those elements of the plan that impact or may be relevant to others must be passed to them by the principal contractor or contractor.	
Inspections	Some temporary works (such as scaffolding and falsework, cofferdams and excavations) are required to have statutory inspections before use and at regular intervals (often before each shift or weekly) after alteration or after any significant adverse event (for example, inspecting a scaffold after high winds).
All such inspections must be carried out by competent persons and a record of the inspections must be retained.	

Other regulations will also apply, such as the Provision and Use of Work Equipment Regulations 1998, the Lifting Operations and Lifting Equipment Regulations 1998 and the Work at Height Regulations 2005.

All of these have inspection requirements.

4.4.2 Contract

Although on multi-contractor sites the principal contractor will have the overall responsibility for temporary works, it is important that all sub-contractors are clear as to what obligations are passed on to others and how sub-contractors are to interact with other sub-contractors and the principal contractor.

Contracts should require the use of good practice approaches (such as British Standards, industry guides, and so on).

For good practice on avoiding ambiguity and disputes, and to ensure high standards are attained, refer to 4.6.1.

The site manager will play an important role by ensuring this interaction occurs in a timely manner, that decisions are recorded and that designers (of permanent or temporary works) are consulted when appropriate.

Although most of this action will be undertaken by the temporary works co-ordinator (TWC) it is important that the site manager maintains an overview and that the interface between site manager and TWC is set out in the company procedures and detailed in the construction phase plan.

4.5 Hazards arising from temporary works

Temporary works produce similar hazards to those within permanent works. Generic hazards for safety and health are shown below.

- Safety.
 - Structural adequacy and integrity.
 - Work at height.
 - Falling objects.
 - Operational management.
- Health.
 - Manual handling.
 - Operational management.

Other hazards will apply on specific projects.

e.g. Operational management failure (safety)

A façade scaffold collapsed in Milton Keynes, causing the death of a worker and injuring several others, because of poor operational site management.

Specifically:

- failure to inspect the scaffold on a regular basis
- insufficient ties
- unauthorised changes
- overloading.

Each of these issues may have been considered small in itself but cumulatively, and over time, they caused a tragic failure.



Poor management of temporary works resulting in a scaffolding collapse

4.6 Good practice

Good practice, drawn up by industry itself, will not only help in the management of temporary works and site safety but also in creating a productive site. Good practice may be divided into **management** and **technical**.

4.6.1 Management

The accepted core document is BS 5975: *Code of Practice for temporary works procedures and the permissible stress design of falsework*. This should be embedded in all contracts to avoid any dispute or uncertainty as to the expected procedures. Main items are shown below.

Designated individual	There should be one person in each organisation who establishes and implements procedures (temporary works design and construction), including appointing competent TWCs. Site managers should be familiar with procedures.
Temporary works co-ordinators (TWC)	Whilst it is not a legal requirement, it is seen as good practice (as outlined in BS 5975), that on each contract the principal contractor or the contractor (on single contractor sites) should appoint one of its own personnel as TWC (known as the principal contractor's TWC (PC's TWC)). It is important that the duties are in writing, accepted, and made known. The TWC is responsible for ensuring that all aspects of the organisation's procedures relating to the management and co-ordination of temporary works are implemented on site. However, if another sub-contractor has control of a specific part of the site and manages their own temporary works, they appoint their own TWC. A schedule of TWC duties is provided in BS 5975. It is preferable that the TWC is not responsible for day-to-day progress of temporary works on site. Where unavoidable, the TWC should ensure that decisions relating to temporary works are safety-based and not influenced by commercial concerns.
Temporary works supervisors (TWS)	On larger or more-complex projects, it may be more desirable to appoint one or more TWS to assist the TWC. It is important that the appointment and duties are in writing, accepted, and made known.
Temporary works register	All sites should have a register of the temporary works. It should include a list of all identified temporary works that will be required throughout the project, giving: ■ a brief description ■ which parties are designing and executing the work ■ significant dates (for example, date design completed and date design checked). The first entry is likely to be site hoarding.
Permits to load/unload	No load-bearing temporary works should be loaded or unloaded without the authorisation of the TWC (in the form of a permit to load or a permit to dismantle).
Information flow	The code stresses the importance of good information flow and co-ordination. Specifically, designers should always have a written brief.

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There are occasions when choosing a solution with greater initial cost may have overall benefits that outweigh the expense. Some examples are shown below.

- Use of prefabricated staircase assemblies to gain vertical access, instead of ladders. Staircases allow quick access and exit in emergencies, multiple usage and are safer for the carrying of tools.
- Use of proprietary barrier systems allows quick installation.
- Formwork with integrated working platforms allows instant, safe and effective access.

4.6.2 Technical

BS 5975 also contains rules for design. Falsework should be designed to these rules or to the European Standards (for example, BS EN 12812) by competent persons with experience in falsework design. Detailed advice is outside the scope of this chapter.

All temporary works should be constructed from drawings and sketches authorised for use by the TWC. Some important watch-points, not requiring explicit technical expertise, are shown below.

- Adequate foundations to falsework (if it doesn't look right, it probably isn't).
- Lateral bracing of adjustable steel props at forkhead level (although some systems and arrangements do not need bracing, it is better to check if none are present).
- All scaffolding requires a formal engineered design, unless it is a basic scaffold, as defined in TG20.
- General standard of construction (such as vertical alignment).
- Adequate protection to those working in excavations.

4.7 Use of equipment and materials

Temporary works often utilise materials and equipment that have been used previously. These should:

- come from a reputable source
- have the previous use authenticated, when relevant
- come with guidance if its safe use is dependent upon the manner in which it is used
- be inspected (usually prior to, and during, delivery) for condition, cracks and distortion, depending upon the component.

The contractor ordering temporary works equipment should have procedures in place to ensure these points are recognised and actioned.

It is particularly important, when using proprietary equipment, to be satisfied that all the components are compatible and are the genuine article.

4.8 What can go wrong?

Most temporary works, following statutory requirements and good practice, are executed without any problem. Nonetheless, there are numerous examples of failure where significant cost, delay and human suffering have resulted. The most common factor of failing is lack of management. This is shown in the table below.

Aspect	Example
General	Insufficient oversight and understanding of the interdependencies between permanent and temporary work packages and the lack of clarity of the responsibilities between sub-contractors. Failure to appoint a TWC or to provide adequate technical back-up.
Change control	Inadequate processes to avoid unauthorised changes (such as removal of scaffold members to allow access for another trade).
Loading	Stacking of materials on working platforms that are only designed for access or light loads.
Time	Insufficient time allowed for the supply chain contractors to undertake their task. This often happens at sub-sub-contract level.
Information	Poor appreciation by the instructing party of the information required by designers and others to deliver a compliant (and safe) output.
Lateral support	Insufficient ties into adjacent structures to provide the necessary lateral support.
Sheeting	Adding sheeting to scaffolds and other above-ground structures (for protection or advertising), which have not been designed to accommodate the additional loading.



Remember, risk thrives at interfaces.

4.9 Briefing and receiving data

4.9.1 Briefing

It is important that those being employed as temporary works contractors or designers are fully briefed (i.e. provided with relevant information, such as design briefs, surveys and investigations). Notwithstanding the requirements of CDM, BS 5975 sets out requirements, as does BS EN 12812 but only for design Class B2. It is important to check that these clauses are sufficient for the particular needs and to appreciate that they do not always apply.



Advice should be sought where the temporary works are beyond the experience of site management.

4.9.2 Receiving data

Equally important is that those employed to design or construct temporary works provide information on any limitations or requirements relating to their work that might impact on others. Examples of the important data required are shown below.

- Pour rate assumed (for vertical formwork).
- Whether the structure may be sheeted.
- Size of tie bolts assumed.
- (Batter) slope inclination for excavations.
- Lateral stability assumptions (such as vertical assemblies relying on the permanent works).

4.10 Generic checklist

It is not possible to produce specific checklists, as temporary works come in a wide range of forms and all are influenced by site-specific situations. However, the following provide an overall reminder.



Is the proposal safety-critical (for example, if it failed could anyone be hurt or have their health adversely affected)?

- If 'Yes', all organisations (and individuals) have statutory responsibilities. Clear procedures are essential.
- If 'No', a problem can nonetheless still cause delay or direct financial loss. Hence, it is a sensible contractual precaution to follow the same guideline principles.



Is there a TWC appointed, acting in accordance with BS 5975?

- If 'Yes', ensure the interface between the TWC and site management is clear and operational. If you are the TWC, ensure you know your obligations and whether you may need to seek assistance.
 - If you are also responsible for site management ensure you are able to exercise the TWC duties independently from other management pressures.
- If 'No', you should refer the matter to the designated individual. The designated individual is the person within the organisation with specific, allocated responsibilities for the setting and implementation of temporary works procedures.



Some small sites can produce substantial risk (such as refurbishment work involving structural alterations, basement construction and deep excavation work).

4.11 Temporary Works forum

The Temporary Works forum (TWf) aims to encourage open discussion of any matter related to temporary works. The group is open to any organisation within the industry and sharing this intent.

The TWf's primary objectives are listed below.

- Give authoritative guidance and, when required, professional leadership to the industry.
- Consider aspects of permanent works and interfaces between permanent works and temporary works as relevant.
- Consider both current practice and future development.
- Be aware of trends and innovations in design, construction and use.



For further information visit the Temporary Works forum website.

4.12 Supporting information

- *Code of Practice for temporary works procedures and the permissible stress design of falsework* (BS 5975:2019).
- *Good practice guidance for tube and fitting scaffolding* (NASC TG20.21).
- *Temporary works – Principles of design and construction* (Thomas Telford, 2019).
- Checklist for erecting and dismantling falsework (The Concrete Society, 2014 (CS123)).
- Checklist for assembly, use and striking of formwork (The Concrete Society, 2014 (CS144)).
- Retention of masonry facades, *Best practice site handbook* (CIRIA, 2003 (C589)).
- *Ground conditions for construction plant* (a good practice guide, Strategic Forum Plant Safety Group, 2014).
- *Management of shoring in excavations*, Part 1, Management process (CPA, 2016 (STIG 13/01)).
- *Management of shoring in excavations*, Part 2, Hazard identification and risk assessment (CPA, 2016 (STIG 1501)).
- Safety in shoring, *The proprietary shoring and piling equipment manual* (CPA, 2006).
- *Temporary works – frequently asked questions*, HSE website.



For further information on the *Preventing catastrophic events in construction* report visit the HSE website.

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Work equipment and hand-held tools

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WORK EQUIPMENT AND HAND-HELD TOOLS

Overview

There are a significant number of fatalities involving work equipment every year. The majority could have been avoided by effective planning and control of equipment use, adequate checking, inspection and maintenance, and thorough training, assessment and briefing of personnel.

This chapter gives an overview of the legal requirements for work equipment and hand-held tools and the steps required to ensure that these obligations are met, so that work equipment can be used safely and efficiently.

5.1 Introduction

All tools and plant used on construction sites are defined as work equipment by the Provision and Use of Work Equipment Regulations 1998 (PUWER), and have the potential to cause harm. Reducing the risk of harm caused by work equipment requires the following steps.

- Assessment of the hazards and risks involved in using work equipment.
- Implementation of measures to eliminate or reduce those risks.
- Ensuring that operators of work equipment are trained, competent and medically fit.
- Ensuring that work equipment is properly maintained and inspected regularly.



For detailed information refer to Chapter C06 Mobile work equipment: 6.7 Training and competence of personnel, and 6.8 Maintenance, checks and inspections.



Problems with work equipment operations often occur because of poor communication when planning the work and briefing personnel.

5.2 Important points

- The term *work equipment* is wide ranging and refers to any item of equipment, machinery or tools being used to carry out work.
- The word *used* means any activity involving the work equipment, including starting, stopping, repairing, modifying, maintaining, servicing or cleaning.
- Examples of work equipment are an excavator, a crane, a hammer, an angle grinder and portable fixing tools.
- All items of work equipment must comply with certain legal requirements.
 - The effort necessary to comply with the requirements will depend upon the complexity of the equipment and its potential to cause harm (for example, a trowel will not require the same level of inspection as a passenger goods hoist).
- The law also requires anyone who uses an item of work equipment to be trained, authorised and competent to do so.
 - The effort and time needed to achieve the required level of competence will depend upon the complexity of the equipment and its potential to cause harm.
- Work equipment must be maintained and inspected as necessary to ensure that it can continue to be used safely.
- Certain types of work equipment must also be subjected to thorough examination, which is generally much more detailed than a routine inspection and can involve dismantling and testing parts of the equipment.

5.3 Legislative requirements

The following is a brief summary of the legislative requirements for plant and work equipment operations.

5.3.1 Provision and Use of Work Equipment Regulations 1998

Under PUWER, *work equipment* means any machinery, appliance, apparatus, tool or installation for use at work (whether exclusively or not). This includes equipment that employees provide for their own use at work. The regulations are concerned with suitability for purpose of work equipment, safeguarding of dangerous parts of machinery, provision of appropriate controls, training of operators and maintenance of all work equipment. PUWER place duties on any person who has control to any extent of any of the following.

- Work equipment.
- A person at work who uses, supervises or manages the use of work equipment.
- The way in which work equipment is used at work (including maintenance).

PUWER apply to employers in respect of work equipment provided for, or used by, their employees, self-employed persons and other persons (such as visitors).

5.3.2 Pressure Systems Safety Regulations 2000

The Pressure Systems Safety Regulations deal with the specific risk arising from both fixed and portable pressure systems. The regulations set out requirements for the following.

- Design and construction.
- Provision of information and marking.
- Installation.
- Use.
- Maintenance, modification and repair.
- Examination.



Safe operating limits should be clearly marked on equipment such as mobile compressors and shot blasting equipment.

5.3.3 Work at Height Regulations 2005

The Work at Height Regulations impose health and safety requirements for equipment that facilitates work at height. Some of these requirements are shown below.

- Steps to be taken to avoid risk from work at height.
- Hierarchy of control for the selection of control measures.
- Organisation and planning, including rescue from height.
- Competence and supervision.
- Selection of work equipment.
- Inspection and testing of work equipment.



Work at height

The regulations define *work at height* as:

- work in any place where, if there were no precautions in place, a person could fall a distance liable to cause personal injury, including at or below ground level
- obtaining access to or egress from such a place while at work, except by a staircase in a permanent workplace, where, if measures required by these regulations were not taken, a person could fall a distance liable to cause personal injury.

5.3.4 Construction (Design and Management) Regulations 2015

The Construction (Design and Management) Regulations place duties on duty holders, including clients, designers and contractors, for the planning, management and monitoring of health, safety and welfare in construction projects and of the co-ordination of the performance of these duties by duty holders. These include a duty on every person working under the control of another to report anything that they think is likely to endanger health or safety. Certain individual regulations have implications for the way in which plant and work equipment operations are carried out.

- There must be safe places of work, including getting to and from the place of work.
- Traffic routes must be suitable for the vehicles that will have to pass over them (for example, dumpers must be used and moved in a way that does not put pedestrians at risk).
- There must be suitable and sufficient lighting for every place of work and traffic route.

5.3.5 Control of Vibration at Work Regulations 2005

These regulations require employers to assess the vibration risk to their employees and eliminate or reduce exposure. Some other employer requirements are listed below.

- Provide information and training to their employees.
- Keep records of risk assessments and control measures.
- Regularly review and update risk assessments.
- Provide suitable health surveillance and keep records.

5.3.6 Control of Noise at Work Regulations 2005

The Control of Noise at Work Regulations require employers to complete the following.

- Assess the risks to their employees from noise at work.
- Take action to reduce the noise exposure that produces those risks.
- Provide their employees with hearing protection if the noise cannot be reduced adequately by using other methods.
- Make sure the legal limits on noise exposure are not exceeded.
- Provide their employees with information, instruction and training.
- Carry out health surveillance where there is a risk to health.

WORK EQUIPMENT AND HAND-HELD TOOLS

5.3.7 Supply of Machinery (Safety) Regulations 2008

The Supply of Machinery (Safety) Regulations are the UK's implementation of European Union Directive 2006/42/EC, the Machinery Directive, which requires that all machinery (including lifting accessories) supplied into the European Union, meets the essential health and safety requirements detailed in Schedule 2 of the regulations. Each machine must be accompanied at the time of supply by an EC declaration of conformity, declaring that the machinery fulfils all the relevant provisions of the regulations.



Plant must be CE marked and meet the essential health and safety requirements of the Supply of Machinery (Safety) Regulations. Consequently, most items of lifting equipment are designed to a harmonised European Standard (such as EN 474-5 for excavators, EN 1459 for telehandlers and EN 12001 for concrete pumps).

5.3.8 British, European and ISO standards

Standards do not generally have the force of law. The application of a standard is almost always voluntary, but compliance with a standard is sometimes quoted in legislation as offering a route to discharging legal obligations. A good example of this is the references to the BS 7121 series in the guidance to the Lifting Operations and Lifting Equipment Regulations 1998 (LOLER).

British Standards are generally restricted to Codes of Practice (CoPs) for safe use of equipment (such as BS 7121-1:2016 *Safe use of cranes*), whilst European (EN) Standards cover requirements for basic principles (Type A), common product requirements (Type B) and specific product requirements (Type C) (such as EN 474-6 *Earth-moving machinery. Safety – Requirements for dumpers*).

International Standards (ISO) cover both the safe use and specification of lifting equipment. They do not have any legal status but are often taken as good practice and are cited as normative references in some EN product standards.



For more detailed information on legislation refer to Section A: Legal and management.

5.4 General work equipment

The building, construction and civil engineering industry uses a vast range of plant and equipment, all of which has to comply with PUWER. It is all referred to as work equipment and includes the following.

- Hand tools (such as hammers and screwdrivers).
- Powered tools, including petrol driven, electric and pneumatic (such as hand drills and circular saws).
- Lifting equipment and accessories (covered under PUWER as well as having requirements under LOLER).
- Testing and laboratory equipment (such as cube crushers).
- Complex structures and machines made up of other components and equipment (such as scaffolds).
- Remote-controlled work equipment and, increasingly, unmanned aerial vehicles (UAVs), often referred to as *drones*.
- Static plant (such as concrete mixers and pumps).



For further information refer to the HSE publication *Safe use of work equipment (L22)*.

5.4.1 Suitability for purpose

All equipment that is used to carry out a work activity must be suitable for that work.

Improvisation is dangerous and can lead to serious accidents (for example, using a hand-held circular saw instead of a router to cut grooves or rebates in timber, or using a dumper to transport persons, creates a dangerous situation).

All work equipment must be regularly maintained to be kept in effective working order and in good repair. It must also be subjected to regular inspection and thorough examination, predetermined by the manufacturers (or scheduled sooner if operated in aggressive environments).

5.4.2 Conformity with UK health, safety and environmental protection requirements

Employers should ensure that any work equipment that is in use or acquired, either new or second-hand, has a CE or UK Conformity Assessment (UKCA) mark. This indicates that it complies with the relevant requirements and has been manufactured to appropriate standards.

The United Kingdom officially left the European Union on 31st December 2020. Before this, products marked with the CE mark were identified as meeting the regulatory standards required, and if they are still in good working order will continue to comply.



UKCA and CE markings

Any equipment purchased from 1st January 2021 within the UK must show the UKCA marking. If imported from the EU, the product must also show the UKCA mark alongside the CE mark.

5.4.3 'Grey' imports

Machines originally supplied from outside the EU frequently do not comply with the requirements of the Machinery Directive and are not CE marked for goods sold in the European Economic Area. If such machines are brought into the EU they cannot be used until they have been modified to meet health, safety and environmental protection requirements. This is a time-consuming and costly process that often negates the benefit of purchasing second-hand machines from outside of the EU.

5.4.4 Mobile plant

This includes excavators, dumpers, forklifts, telehandlers, rollers, mobile elevating work platforms (MEWPs) and mobile cranes. Mobile plant may either be controlled by the operator riding on the machine or they may be pedestrian controlled. Increasingly, some mobile plant is available as specialist, remote controlled equipment.



For further information on mobile plant and remote controlled equipment refer to Chapter C06 Mobile work equipment.

5.4.5 Static plant

This includes items such as construction hoists (*refer to Chapter C08 Lifting equipment*), compressors and pumps (*refer to 5.8 and 5.9*), batching plants, concrete mixers, and rebar benders and cutters.

5.4.6 Lifting equipment

Both mobile and static plant may be designed as lifting equipment. This includes mobile cranes, tower cranes, MEWPs, hoists, telehandlers and excavators used for lifting.



For further information refer to Chapter C07 Lifting operations and Chapter C08 Lifting equipment.

5.4.7 Manually operated hand-held power tools and work equipment

Selection of the correct equipment for the task is vitally important. Choosing modern equipment that reduces the risk of injury or ill health is fundamental to creating and improving safe working environments.

The environment in which the equipment is used also has a bearing on the safety of the operative. In certain circumstances it may be safe to use hand tools in a confined space, but to use powered equivalents that create dust and fumes could have fatal results.

Hand-held power tools, whether powered electrically, by internal combustion engine, hydraulically, by cartridge or by compressed air, are covered by the definition.

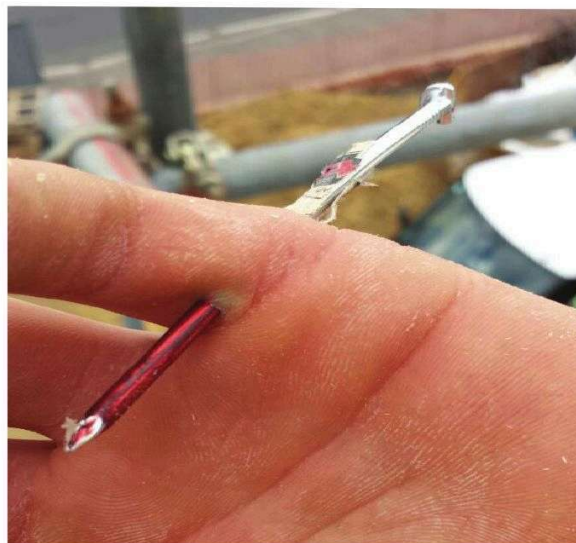
As the use of these types of tool has increased, so has the potential for serious injury. Old type, hand-operated smoothing planes could cause serious cuts if incorrectly used, but those injuries bear no comparison with the injuries that could occur when incorrectly using their modern electrically powered equivalent.

Cutting chases in a wall using an older type cutting disc produces unacceptable amounts of dust. Modern equipment that is fitted with a dust suppression mechanism is safer and much healthier.

Also included within this category is the type of equipment that requires human effort in its operation (such as jacks, bar-bending machines, pipe-threaders and other machines where the principles of leverage or torque are utilised).

Many jobs (such as nailing slate battens on a roof), which traditionally would have been completed by using a hammer, are now carried out using nail guns powered by gas canisters.

Equipment of this type is open to abuse, which can result in horrific consequences. It is possible to propel a nail in free flight from a nail gun, giving it the lethal potential of a firearm (*for further information refer to 5.6 Cartridge and gas-operated fixing tools*).



A lack of training and/or supervision, concentration and poor practice can result in serious injury

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5.4.8 Hand tools

The general requirements of PUWER apply as much to hand tools as they do to more complex items of work equipment. Unfortunately, this is often not appreciated by those managing hand tools, who tend to only deal with problems when they occur, rather than eliminating hazards in the first place.

Two common hazards are listed below.

1. Cold chisels with burred-over mushroom heads can result in a serious eye injury that may result in the loss of an eye if a burr shatters and breaks off during use.
2. A hammer head becoming detached from its shaft can cause a serious injury to the user or a person nearby.

In both examples, the potential to cause harm can be substantially reduced if a system of regular checks, inspection and maintenance, as required by the regulations, is instigated and followed.

5.4.9 Non-mechanical access equipment

5.4.9.1 Scaffolds

If incorrectly erected or misused, scaffolds can be the cause of accidents. Therefore, they must meet the following requirements.

- Be suitably designed.
- Use suitable components that have been inspected prior to use.
- Be inspected every seven days, after alteration, and after any unexpected occurrence (such as impact damage or high winds), the results of the inspection recorded, and proactive maintenance regimes put into place.
- Not be interfered with or altered by unauthorised persons.

Many accidents have occurred due to the installation of scaffold with damaged equipment (such as split boards and bent tubes).



For further information on access scaffolding refer to Chapter D04 Scaffolding.

5.4.9.2 Ladders and other wooden access equipment

HSE guidance is that ladders and stepladders should be used only as a place of work when the nature of the work meets the following criteria.

- It is of short duration (up to a maximum of 30 minutes).
- It is of a light (up to 10 kg) nature, requiring no heavy lifting, carrying or a destabilising pressure applied by the user or equipment in carrying out the work (i.e. minimal manual handling).
- It allows one hand to be available at all times for holding on to the ladder or stepladder.
- It requires nothing to be carried that would cause instability of the ladder, stepladder or user.

Additionally, when stepladders are used as a place of work they should be positioned so that the user faces the work as the stepladder is climbed. A stepladder must not be positioned so that the user is side-on to the work, where the nature of the work would apply a sideways pressure and cause the stepladder to become unstable.



For further information on the selection and use of ladders and stepladders refer to Chapter D03 Common access equipment.



For further guidance refer to the HSE leaflet *Safe use of ladders and stepladders (INDG455)* and the Ladder Association website.

The following are example of **types of ladder** that are available.

Standing ladders are a single stage ladder up to 6 m in length.

Pole ladders are a variant of standing ladders, but with the stiles having been made from a long, wooden pole cut centrally to give even strength and flexibility. Lengths can vary up to a maximum of 10 m.

The practice of shortening a pole ladder to fit a particular situation should be discouraged. Care must be taken to ensure that the fabric of the ladder remains stable if a tie wire is removed.

Extension ladders consist of two or three sections coupled together and extended by sliding over or inside each other. Longer multi-stage ladders are extended by means of a rope and pulley. The maximum reach of a single run ladder is 9 m. At this height and above, resting points must be provided.

Stepladders are of various types, have flat rectangular treads and are usually free standing.

Platform stepladders have a built-in working platform, and newer models can have wheels for ease of movement, handrails and restraint chains. Due to the provision of a platform, these are usually a better option for work that is regularly carried out at the same height.

Roof ladders are ladders with a hook on the top end for securing over the ridge of a roof.

Aluminium and steel ladders are available in various types. Their main advantage is that they are light and weather resistant. They can be prone to slipping at the base if the rubber feet are not properly maintained.

Telescopic ladders are a variation of leaning ladders, but they do not all work the same way. They should always be used, stored and transported with care and kept clean, and is always important you read and follow the user instructions provided by the manufacturer.



Portable ladders, steps, trestles and lightweight staging are covered by BS 1129:1990, BS 2037:1994 and BS EN 131.



The Ladder Association, a not-for-profit body dedicated to promoting safe ladder use, has produced a safety guide for users of telescopic ladders – an increasingly popular piece of equipment for working at height – following recent reports of unsafe products in the UK. Visit the Ladder Association website to download a free copy of the guidance.

In November 2017, the British Standards and regulations for ladders and steps changed. BS 2037 Class 1 Industrial and Class 3 Domestic standards were withdrawn to create two European-wide categories: **EN 131 Trade and Industrial** – heavy duty and industrial use (for professional users) and **EN 131 Domestic** (for non-professional users; not suitable for construction sites).

The new standards will ensure products are built with additional safety standards, such as wider bases and stabilisers on ladders over 3 m in height.

5.5 Abrasive wheels

Abrasive wheel machines are potentially dangerous if not used correctly by competent persons. Most accidents result from selecting the wrong type of wheel or from over-speeding. It is essential that:

- the right abrasive wheel for the job is chosen
- it is run at the correct speed and guards are fitted
- the wheel is correctly mounted by a competent person on a compatible machine
- high-impact eye protection is used.

In most circumstances abrasive wheels rotate at high speeds and contact with the revolving wheel can cause serious injury.

When using a grinding or cutting wheel, a stream of hot, abrasive particles is thrown off, which can cause injury, particularly to the eyes. Finally, there is always a risk of the wheel disintegrating or bursting as it revolves.

Fragments of the wheel can be projected a great distance, at great speed and in all directions.

5.5.1 Types of abrasive wheel

An abrasive wheel is generally defined as a power-driven wheel, cylinder, disc or point having abrasive particles. It may consist entirely of abrasive particles, or be of metal, wood, cloth, felt, rubber or paper, with a surface covered with abrasive material.

It may also be formed as a ring or segments of abrasive materials. These types of abrasive wheel are used for shaping material via grinding operations. Other types of abrasive wheel are those used for cutting rather than grinding operations. These are circular metal blades, usually with diamond impregnated tips, used for cutting through concrete, steel, and so on.

5.5.2 Choosing the correct wheel

The majority of accidents involving abrasive wheels are the result of using the wrong type of wheel for the job in hand. Had the manufacturer's recommendations been complied with then these accidents would have been avoided.

Abrasive wheels should comply with the requirements of BS EN 12413 and BS ISO 525.

Many types and grades of abrasive wheel are available, and the correct selection is important. As a general rule, soft grade wheels are most suitable for use on hard materials and hard grade wheels are for use on soft materials. Coarse grains are for the rapid removal of material and fine grains are for polishing.

The use of an unsuitable wheel may result in the wheel face becoming loaded as the pores are clogged by the material being removed. If the wheel is too hard or too fine it may become glazed or polished.

The consequence, in both cases, may be that the operative will press too hard, in an attempt to get the work done, and cause the wheel to break. It is important that only reinforced, resin-bonded abrasive wheels should be used with portable grinding machines.

Abrasive wheels should be marked to conform to Annex A of BS EN 12413.



For further guidance refer to the HSE publication *Safety in the use of abrasive wheels* (HSG17).

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5.5.3 Maximum permissible speed

The maximum permissible speed in revolutions per minute (rpm) and metres per second (m/s) specified by manufacturers should be marked on every abrasive wheel larger than 80 mm in diameter, or on the blotter or identification label that is sometimes attached to it. Since it is not practicable to mark smaller wheels, the maximum permissible speed in rpm of wheels 80 mm in diameter or less should be stated in a notice posted in a position where it can easily be read. For speeds of 50 m/s and above, colour coded stripes will appear on the wheel.

Metres per second	Colour
50 m/s	Blue
60 m/s	Yellow
80 m/s	Red
100 m/s	Green
125 m/s	Yellow/blue

5.5.4 Restrictions of use

Annex A of BS EN 12413 and BS ISO 525 specify how wheels should be marked to indicate specific restrictions for use.

- RE1: Not permitted for hand-held and manually guided grinding.
- RE2: Not permitted for hand-held cutting-off machines.
- RE3: Not suitable for wet grinding.
- RE4: Only permitted for totally enclosed working areas.
- RE6: Not permitted for face grinding.

5.5.5 Shelf life

All organic bonded wheels for hand-held applications will have a use-by date, which is generally three years from the date of manufacture. The organic bond will break down over a period of time, increasing the likelihood of wheel burst during normal use.

5.5.6 Traceable number

A code number should be marked on the wheel to indicate the source and manufacturing details of the wheel.

5.5.7 Wheels bursting

The two main causes of abrasive wheels bursting are over-speeding and the incorrect mounting of the wheel.

5.5.7.1 Over-speeding

Abrasive wheels must not be run in excess of the maximum permissible speed recommended by the manufacturer. This should be clearly marked in rpm on all wheels over 80 mm (approximately three inches) in diameter.

For smaller wheels, a notice stating the maximum speed permissible should be displayed on or adjacent to where the machine is being operated. Any type of record may be kept, from computer database to a written record. The only restrictions are that a hard copy of the information must be available on request.

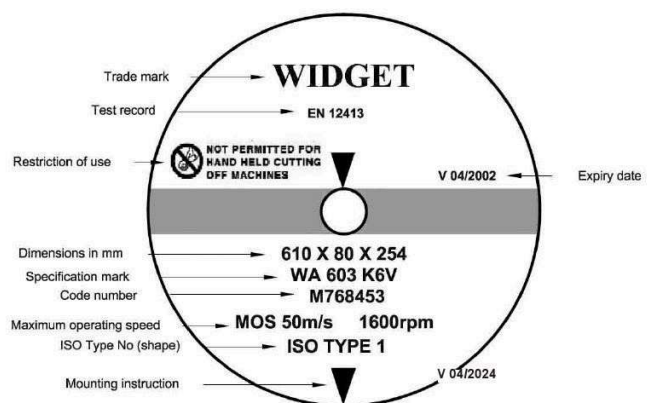
Speed of spindle. Every power-driven machine for use with abrasive wheels should have a notice fixed to it, specifying the maximum and minimum spindle speeds. Machines that are not marked with their spindle speed must **not** be used.

Sometimes these markings are very small. If this is the case, consideration should be given to painting or engraving the spindle speed clearly on the machine. Always keep the markings clean and in good condition so that they are easy to read.

Maximum spindle speeds must never be exceeded. Governors, or other devices to control or limit speed, must be properly maintained and in good working order.

The maximum spindle speed on equipment should be checked on a regular basis. This is particularly important on engine driven machines where the speed can change over a period of time as a result of component wear.

Peripheral speeds. Care must be taken to operate the wheel at the optimum speed that is recommended by the manufacturer.



Abrasive wheel marking



Spindle speeds can be checked by using a stroboscopic light meter

As an abrasive wheel wears down, its peripheral speed (at the edge of the wheel) is reduced and this, in turn, reduces its efficiency. A wheel operated at too low a speed will not cut properly and there will be a tendency to press the wheel hard against the work piece. This may damage the wheel and cause it to burst.

5.5.7.2 Mounting of wheels

In order to be able to demonstrate compliance with legislation, abrasive wheels should only be mounted by a trained, competent and authorised person appointed to this duty by their employer.



Never attempt to mount a wheel onto a machine unless you are authorised, trained and competent to do so.

5.5.8 Training

Training for the mounting of abrasive wheels must **only** be given by a competent person, which often involves attending an external course. It must cover the grades, types and marking of wheels, the use, hazards, speeds, storage, handling, inspection, testing, dressing, adjustments and the functions of associated components (such as flanges, blotters, bushes and locking nuts).



Details of suitable training courses and training providers can be found on the British Abrasives Federation website.

5.5.9 Maintenance

In addition to checks immediately before and after each use, every machine should be regularly serviced by a competent person in accordance with manufacturer instructions. A maintenance record should be kept. It is good practice to record the type and serial number of all abrasive wheel machines, together with the date each machine is due for service. When servicing has been carried out, ensure that details of any completed work are entered, together with the date that the next service is due. All records should be kept up-to-date.

5.5.10 Diamond blades

A diamond blade is a circular steel disc with a diamond bearing edge. The edge of the blade may be smooth or textured, with a continuous rim, or a segmented rim with smaller, individual sections. The blade core is a precision-made steel disc and may have a continuous or slotted rim. The slots (also called gullets) provide faster cooling by allowing water or air to flow between the segments. The slots also allow the blade to flex under cutting pressure.

As with bonded abrasive wheels, people working with diamond blades must be properly trained and competent. Most of the safe procedures and precautions for using bonded abrasive wheels on hand-held machines apply to diamond blades. Additional advice and information can be obtained from suppliers and manufacturers. Diamond blades are **directional** and must be mounted so that they rotate in the direction marked on the blade. Different wheels are designed to cut different materials, a requirement that must be strictly adhered to.



Diamond blades can significantly deteriorate and become unsafe if they are used to cut the wrong type of material.



Use the correct equipment for the task

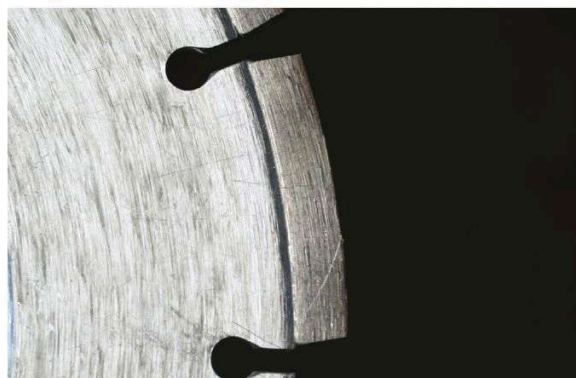
A worker used a concrete blade to cut a series of post holes through an asphalt surface to receive concrete fence posts. As a result, the blade failed and several 50 mm long segment blade tips broke off at high speed, two of which embedded deep within his knee cap. He suffered high blood loss, was airlifted to hospital and the tips were removed during reconstructive surgery. He was off work for over six months and never fully recovered. Providing an additional asphalt blade would have avoided the accident.

Diamond blades can be used on hand-held machines at peripheral speeds of up to 100 m/s, provided that the machine is designed for this speed and the marked maximum permissible operating speed of the blade is not exceeded. Under no circumstances must the blade be allowed to overheat.

Periodically, it may be necessary to redress (sharpen) the blade by making cuts in a special dressing block or abrasive block. These wheels can last a comparatively long time. Before use and with the power disconnected, make a special point of checking the condition of the spindle bearings, as these can deteriorate considerably during the lifetime of a diamond cutting wheel. Pre-use checks for wheel defects should be carried out before each use.



Reputable manufacturers will provide inspection guides for their blades.



Diamond blades need to be inspected on a regular basis for defects

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5.5.11 In-use considerations

5.5.11.1 Guards (fixed and hand-held machines)

- The guard should be designed, and be sufficiently robust, to contain any fragments in the event of the wheel bursting.
- The guard must be securely attached to the body of the machine, properly adjusted and maintained.
- The guard or shield supplied by the manufacturer must be adjusted to permit exposure of the minimum amount of wheel or disc to enable the job to be carried out. It must not be left off or altered in any way.

5.5.11.2 Defects

Employees are required to report any defects in work equipment, including abrasive wheels, guards or tool rests, to their employer.

5.5.11.3 Tool rests

Rests should be secured and adjusted so that they are as close as practicable to the wheel and no more than 3 mm away. If the gap between the wheel and the rest is too great, the work piece may become trapped and an accident may result.



Continue to display cautionary notices concerning the dangers of abrasive wheels, wherever abrasive wheels are used or changed. This also serves as a useful reminder to employees of the training they have received.

5.5.12 Health considerations

5.5.12.1 Respiratory protection

Depending upon what material is being cut, it is possible that the user of an abrasive wheel may be vulnerable to inhaling hazardous dust. With fixed machinery (such as bench grinders or fixed concrete saws), it may be possible to reduce airborne dust to a safer level by installing a fixed local exhaust ventilation system.

The following would be more applicable to portable machines.

- Purchasing abrasive wheel machines that incorporate a facility for wet cutting.
- The attachment of a standalone dust extraction/collection unit (on-tool extraction).

It is not always reasonably practicable to completely eliminate respiratory hazards. If, after applying all other measures, there is a residual likelihood of exposure, suitable respiratory protective equipment (RPE) should be provided to each person exposed. Therefore, the operator in both examples listed above would need to be issued with appropriate RPE and also be face-fit tested.



A petrol-driven disc cutter with the facility for wet cutting



For further information refer to Chapter B06 Personal protective equipment.

5.5.12.2 Vibration

The very nature of grinding or cutting work using abrasive wheels is likely to subject the operator to a degree of hand-arm vibration. In extreme cases, this can lead to permanent and disabling injuries.

Employers must assess the risk and exposure levels to employees who carry out such work, and put preventative and protective measures in place to ensure that employee health is not adversely affected.



For further information on control of vibration refer to Chapter B12 Vibration.

5.5.12.3 Noise

Grinding and cutting operations using abrasive wheel machines will inevitably result in an increased level of noise, both from the machine itself (for example, a petrol-driven disc cutter) and from contact between the cutting disc or grinding wheel and the material being worked upon. In many cases, the level of noise generated will exceed the upper exposure action value, meaning that, unless other control measures can be put in place, personal hearing protection must be made available and worn. Depending upon the level of noise and the proximity of other people, it may be necessary for them also to wear hearing protection.



For further information on control of noise refer to Chapter B11 Noise.

5.5.13 Safety considerations

During cutting or grinding processes, a stream of dust or abrasive particles and hot sparks is thrown off. These particles could be travelling at speeds in excess of 170 mph and can cause serious injury to the eyes.

Because there is an obvious risk of eye injuries when an abrasive wheel is being used, **high-impact resistant** eye protection should be provided in accordance with the Personal Protective Equipment Regulations (goggles, a face shield or visor, to BS EN 166).



For further information on eye protection refer to Chapter B06 Personal protective equipment.

- The machine must have an efficient starting and stopping device that is easy to access and operate.
- The floor area of the workplace must be kept in good condition, free of loose material and trip hazards, and should not be slippery.
- Abrasive wheels should be properly stored, flat and preferably in their boxes, and in accordance with the manufacturer's guidance. Care must be taken to ensure that any manuals are retained so that the manufacturer's guidance and instructions can be read. Labels on the discs themselves should be kept clean and undamaged so that the information can be easily read.



For detailed information on abrasive wheels refer to the HSE publication *Safety in the use of abrasive wheels* (HSG17).

5.6 Cartridge and gas-operated fixing tools

Portable fixing tools use the power of an explosive charge or a gas propellant (from cartridges or a gas canister) to drive a fixing device into position in a base material. They are particularly useful if there are a large number of repetitive fixings to be made, where a portable fixing tool reduces the time and labour expended in this area. However, the resemblance of the tools to the shape and action of a gun can lead to their misuse, especially by young and inexperienced workers.

They can be extremely dangerous if used incorrectly. Operatives must be trained, competent and authorised and be of a sufficiently mature and responsible disposition.

Poor technique, or the use of incorrect equipment, will result in poor or defective fixing. The tool, type of cartridge or fuel cell, type of fixing and the base material must all be compatible.

In most cases, items of additional personal protective equipment (PPE) (such as hearing, respiratory and eye protection) will have to be worn. At the end of the job all unused cartridges must be accounted for.



It should always be remembered that portable fixing tools are potentially lethal if they are used recklessly or incompetently.

5.6.1 Cartridge-operated tools

A cartridge-operated tool works by using a firing cap or cartridge to provide propellant to drive a fixing home. These tools must not be used in areas where a flammable atmosphere or risk of dust explosion may exist. Listed below are the primary factors that will ensure the proper and safe use of cartridge-operated tools.

- Adequate information, instruction, training and supervision.
- Competent and responsible users, who are over 18 years old and are not colour blind (cartridges are colour coded for different levels of power – see 5.6.1.4).
- The compatibility of the base material, the type of fixing and the cartridge strength.
- Restricting access to the work area during fixing activities.
- The provision and use of appropriate PPE.
- Carrying out work in accordance with *Code of Practice for safe use of powder actuated fixing systems* (BS 4078-1).
- Using cartridge-operated tools that comply with *Specification for tools* (BS 4078-2).



High-powered cartridge-operated tools can operate at the same velocity as a small firearm.

5.6.1.1 Types of tool

There are two main types of cartridge-operated tool.

Indirect-acting, where the driving force is transmitted to the fixing by means of the expanding explosive gas acting on a piston. These tend to be low power.

Direct-acting, where the explosive force of the cartridge acts directly on the fixing, driving it along the barrel into the background material. These tend to be high power.

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5.6.1.2 Power level of tools

Cartridge tools are generally classed as high power and low power.

Low power is defined as giving the pin a kinetic energy not greater than 3.5 m/kg/f and a velocity not greater than 98.5 m per sec.

High power applies to any values greater than those above. Using high-power tools can result in dangerous through-shoots. This is where the fixing is fired right through the background material.

Most of the commonly available tools are low power and indirect acting. These are by far the safest. There are high-power tools for special applications, and some old, high-power (direct-acting) tools are still in use.

Hammer-activated tools are nearly always low power.

Some modern tools have provision for varying the power level (within the low-power range) by means of an adjustment that changes the size of the gas expansion chamber. There are also interchangeable pistons for different fixings or depths of penetration. Tools incorporating these features require a smaller range of cartridges.

5.6.1.3 Safety devices

All tools should incorporate a contact pressure safety device, which prevents them being fired unless the muzzle is pressed hard against the workface. They should always incorporate a drop-firing safety device that prevents the tool from firing when it is dropped onto a hard surface.

In addition, some tools are also equipped with an unintentional firing safety device. This prevents the tool from firing if the trigger is pulled before the tool is pressed against the work surface.

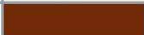
It should **only** be possible to fire the tool when it is correctly pressed against the material being fixed.

5.6.1.4 Cartridges

Any premises or any place where the cartridges for a cartridge tool are stored should be secure, dry and cool, and may need to be licensed. Checks should be made with the local police station or Local Authority as to whether this is necessary and, if so, where such a licence can be obtained.

Cartridges are designed for specific brands or types of tool and are not interchangeable, even if they are of a similar type or appearance. They are available in different strengths. It is preferable to start with a lower strength cartridge for a test fixing and then change to a more powerful cartridge if the depth of penetration is not sufficient.

BS 4078 requires the strength of the cartridges to be marked on the packaging and each cartridge to be colour coded to indicate its strength. The colour codes set out in BS 4078 are shown on the right.

Cartridge strength	Colour	
Extra low (XL)		Brown
Low (L)		Green
Low/medium (LM)		Yellow
Medium (M)		Blue
Medium/high (MH)		Red
High (H)		White
Extra high (XH)		Black



For further information refer to **Cartridge operated hand-held tools. Safety requirements. Fixing and hard marking tools (BS EN 15895).**

It should be noted that the code is not universally followed, and that colour coding alone must never be relied on as an indicator of the cartridge strength. Cartridges should be retained in the packaging (which identifies their strength) and not carried loose.

Both cartridges and fixing nails are now available on plastic strips.



Controls for dispensed and unused cartridges

Proper controls must be put in place to account for all dispensed and unused cartridges.

Fully dispensed cartridges in single or strip form are not classed as hazardous waste, and can be disposed of in accordance with normal rules of non-hazardous waste.

Unused or unserviceable cartridges that cannot be fired are hazardous, and in **no circumstances** should be exposed to fire or heat, or be subject to mechanical impact.

For the purpose of waste classification, unused cartridges often have a hazardous European waste code entry of 16 04 01* *Waste ammunition*, but this should be checked in each case from the safety data provided by the supplier of the cartridge.

The supplier should be contacted to discuss suitable arrangements for the return/disposal of any unused or unserviceable cartridges.

Cartridges should be returned to their original packaging so that they can be properly identified.

5.6.1.5 Suitability of base materials

Attempting to fix into unsuitable materials with cartridge tools is dangerous. Before firing the first fixing, a simple test should be made by driving a fixing of the intended type into the base material with a hammer. The result will show whether the material is suitable. No attempt should be made to fix into unsuitable materials. The table below provides general guidance.

Material	Result	Conclusion
Plaster, plywood and lightweight blocks	Sinks in easily	Too soft (not suitable)
Marble, some rock, hardened steel and weld metal	Fixing blunted	Too hard (not suitable)
Glass, glazed tiles, slates and some cast-iron	Material cracks or shatters	Too brittle (not suitable)
Sound wood, concrete and mild steel	Clear impression of fastener point	Suitable

5.6.1.6 Hazards in use

Hazards from the use of cartridge tools generally arise from one or more of the following three factors.

1. Lack of competence, knowledge or training.
2. Misuse, whether deliberate or due to ignorance of proper use.
3. Poor maintenance, rendering the equipment defective or unsafe.

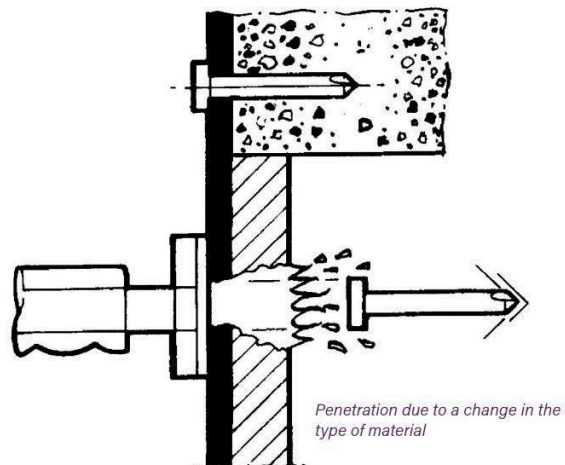
Two conditions that specifically need to be guarded against are **through-penetration**, where the fixing goes through the material emerging in free flight on the other side, and **ricochet**, where the fixing and debris is deflected after firing, possibly towards the operator.

5.6.1.7 Causes of through-penetration

- Cartridge too powerful for the material being fixed to.
- Fixing into voids in the structure.
- The background material being fixed into is too thin.
- Changes in type or consistency of the material penetrated.
- Not establishing the density of the material being fixed into.

To avoid these hazards:

- check the suitability of the material for cartridge-fired fixing
- if necessary, make a trial fixing using a low-powered cartridge
- check and guard the area behind the material or structure into which the fixing is being fired, to prevent access of unauthorised persons
- use an indirect (lower powered) acting tool.

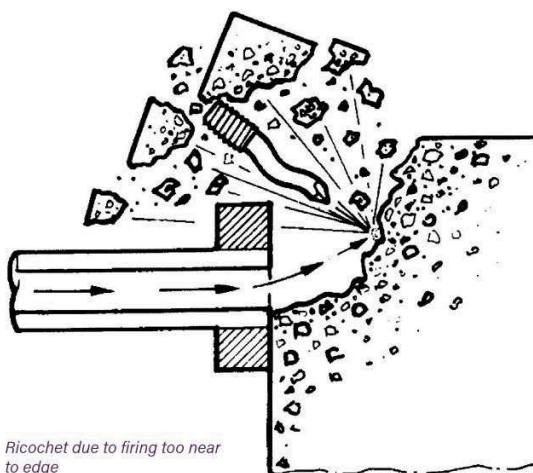


5.6.1.8 Causes of ricochet

- Firing into the hole of a previously attempted fixing.
- Attempting to fix into excessively hard materials (such as hardened steel or welded areas).
- Cartridge tools not held square onto the work surface, causing the pin to strike at an angle and then be deflected.
- Attempting to fix too near to an edge.
- Hitting a reinforcing rod or dense aggregate hidden just under the surface.

To avoid ricochets:

- fixings should be at the recommended distance (or greater) from failed attempts
- do not fix into unfamiliar background materials without first safely checking their suitability for cartridge fixing
- tools should be at right angles to the work surface. The whole rim of the splinter guard should be firmly placed against the workface so as to stabilise the tool and not leave gaps.



The risk of ricochet is reduced by the use of low-powered, indirect acting tools. If high-powered tools are used, there is a risk that the fixing can be deflected and come back towards the operator.

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5.6.1.9 General precautions

Recoil of a cartridge tool can lead to loss of balance, which is made worse if working from an unstable workplace. In this instance, only low-power tools should be used. Operators not familiar with cartridge tools, should test the tool for recoil before use. In all circumstances, an adequate and safe footing is a necessity. Makeshift platforms should not be used.

In the event of a **misfire**, the cartridge tool should be kept pressed in position against the workface for at least **30 seconds**, to allow for any delayed detonation. Following this time period, the cartridge must then be removed strictly in accordance with the manufacturer's instructions. Cartridges that have misfired should be stored in a metal box and returned to the supplier.



Misfired cartridges should not be removed from the tool by levering under the rim. Some types are rim detonated and could be fired by this action. Only the proper extraction tool, as supplied by the manufacturer, should be used.

Loading of tools should be carried out immediately prior to use. Once the tool is loaded, it must never be pointed towards other people or at any part of the operator's body. Unused cartridges must be removed from the tool as soon as possible.

Storage and issue of tools. Only responsible and competent persons should supervise and check the acquisition, issue, use, return and maintenance of cartridge-operated tools.

Cartridge-operated tools and cartridges should be stored in a place that is secure, dry and cool. The issue of such tools and cartridges should be strictly controlled, and licensed if necessary. The following points should be clearly noted and understood.

- Cartridge tools should only be stored in an unloaded state.
- Different strengths of cartridge should be clearly identified and kept separate.
- The use of different makes of equipment on one site should be limited as far as possible.
- The manufacturer's instructions on the safe use of the cartridge tool should always be made available.

Personal protective equipment, complying with the relevant British Standards, should be used by operatives using cartridge-operated and gas-operated tools.

- High-impact resistant eye protection to the applicable BS EN 166 impact rating must be provided and worn at all times when handling and operating cartridge-operated tools, their cartridges or gas-operated tools.
- Noise levels will vary with the make of tool but all tools create a high intensity, short duration noise, so suitable hearing protection should be worn.
- Safety helmets to BS EN 397 should be worn while cartridge tools are in use.

The safety of other people in the vicinity of where cartridge-operated or gas-operated tools are being used, is a factor which should have been covered in the risk assessment, must be considered and all necessary precautions taken.



For further information refer to Chapter B06 Personal protective equipment.

5.6.1.10 Safe battery charging

Good practice when charging the lithium-ion batteries of hand-held tools and equipment, in order to reduce the risk of potential fire risk, would be to:

- Never charge batteries near flammable materials or objects.
- Never charge batteries in places where high temperatures or exposure to direct sunlight may be expected.
- Never cover the batteries when charging.
- Ensure that overheat protection is installed at the charging socket.
- Never overcharge or leave to charge over night on a trickle charge; use time-out protection.

5.6.2 Gas-powered tools

Lighter duty gas-powered tools can be used for the fixing of timber and other low density materials. These fixing tools are generally lighter in weight and use a battery and fuel cell, either propane or other gas, to act as a propellant rather than a cartridge. The fuel cell, typically a small aerosol canister, is mounted within the tool. The safety issues that need to be considered when using a gas-powered tool are shown below.

- Operators must be trained. Usually this can be provided by the supplier of the equipment. Operators must also be in possession of, and be familiar with, the relevant equipment instructions.
- When firing, the operator must be in a stable position and holding the tool at right angles to the work.
- Firing must not take place when others are behind the work. As with cartridge tools, through-penetration can occur.
- If a malfunction occurs the tool can usually be fired again (unlike when using a cartridge tool). However, the manufacturer's instruction manual should be consulted.

- The tool and the fuel cells must not be exposed to extremes of heat or used within a flammable atmosphere.
- Harmful fumes are expelled during use and therefore the tool should be used in a well-ventilated area.
- Because of the mode of operation and speed of use, the tool can become hot. Reference should be made to warnings of hot areas on the tool.
- Fuel cells must not be damaged, pierced, punctured, burnt or opened, even after use.
- Fuel cells should be disposed of in accordance with local regulations for aerosol products.

5.7 Woodworking machinery

- Woodworking machines are classified as work equipment and must therefore comply with certain legal requirements.
- Many items of woodworking equipment, with their partially exposed blades or cutters, have the potential to cause serious personal injury if they are not properly used and maintained.
- Operators of woodworking machinery must be trained on each type of machine that they are required to operate and be judged as competent by their employer.
- Hand-held, bench-mounted and free-standing woodworking machines of all types present particular hazards to the operator and, in some cases, other people.
- The use of woodworking machines has the potential to cause occupational health problems if appropriate preventative measures are not taken to address the risk from dust, vibration and noise.

Several hundred accidents on different types of woodworking machinery are reported annually to HSE.

Of these, approximately 40% occur on circular saws and 25% on planing machines.

Woodworking machine cutters can inflict serious injuries and it is essential that the regulations for guarding them are strictly observed.

Neglect or ignorance of the regulations governing the use of such machinery creates the conditions in which accidents occur.

5.7.1 Types of machine

The following non-exhaustive list gives examples of different types of woodworking machine.

- Circular saws.
- Sanding machines.
- Bandsaws.
- Routing machines.
- Planing/thicknessing machines.
- Chainsaws.
- Mortising machines.
- Tenoning machines.
- Vertical spindle moulding machines, including high-speed routing machines.
- Multi-cutter moulding machines having two or more cutter spindles.
- Trenching machines.
- Boring machines.
- Automatic and semi-automatic lathes.

5.7.2 Health and safety risks

Some examples of occupational health problems caused by using woodworking machines are listed below.



- **Noise-induced hearing loss, caused by exposure to noisy environments.**
- **Hand-arm vibration syndrome, caused by vibration of equipment.**
- **Chronic respiratory diseases, caused by the inhalation of fine dust particles.**
- **Eye injury.**
- **Dermatitis, caused by contact with fine wood dust.**

WORK EQUIPMENT AND HAND-HELD TOOLS

5.7.2.1 Noise-induced hearing loss

Woodworking machines can be particularly noisy when in use and the noise generated will be a significant hazard to health unless adequately controlled. Ideally, the level of noise will be controlled by means other than issuing PPE (for example, using screening or anti-vibration mountings).



For further information on control of noise refer to Chapter B11 Noise.

5.7.2.2 Hand-arm vibration syndrome

In the context of using woodworking machines, the predominant problem is considered to be hand-arm vibration. The necessary training and on-going health surveillance should be implemented to prevent occurrence.

5.7.2.3 Respiratory problems

Woodworking machines inevitably produce dust that may affect the operator and persons in the vicinity. Dust should be treated and assessed in accordance with the Control of Substances Hazardous to Health Regulations 2002. It should be controlled by effective extraction, collection systems or using dampening techniques in addition to wearing **face-fitted** RPE. Fumes can also be generated from adhesives and substances used for the treatment or finishes of the wood.



For further information on control of dust refer to Chapter B10 Dust and fumes (Respiratory hazards).

5.7.2.4 Eye injury

Depending on the machine being used and the nature of the work being carried out, it may be necessary for machine operators (and anyone assisting them) to wear eye protection (such as goggles) or a full-face shield of a suitable impact-resistant grade.

5.7.2.5 Dermatitis

It may be necessary to provide operatives with suitable gloves and protective creams to protect against skin damage whilst handling woods, particularly unplanned timber. Furthermore, the skin of a small number of persons can be adversely affected by the natural oils that occur in some woods, notably cedar.



For further information refer to Chapter B06 Personal protective equipment.

5.7.3 Controls

Every machine must be fitted with start and stop controls that can be quickly and easily accessed by the operator.

5.7.4 Braking

An essential safety feature of most woodworking machinery is the provision of an automatic brake. This is to ensure that, if there is a risk of an employee coming into contact with the cutters during the rundown period, the machine stops within a period defined by CEN European standards.

5.7.5 Guarding cutters

Cutters include saw blades, chain cutters, knives, boring tools, detachable cutters and solid cutters. These must be guarded to the greatest practicable extent, having regard to the work being done. Guards must be of substantial construction, properly secured and adjusted, and constantly in position while cutters are in motion.

Adjustments **must not** be made to any guard whilst the cutters are in motion, unless safe means (that is mechanical adjusters) are provided for those adjustments to be made. Every dangerous part of any woodworking machine must be securely fenced, or be in such a position or of such construction that it is as safe to every person on the site as it would be if it were fenced.

5.7.6 Portable, electrically powered saws

Portable, electrically powered saws must always be disconnected from the electrical supply before any adjustments, repairs, cleaning or maintenance is carried out.

Whenever possible, to reduce the risk from electric shock, all portable, electric tools should be battery-powered or 110 volts and, if 110 volt supply, be earthed or double insulated. The mains input to a 110 volt transformer should be protected by a residual-current device (RCD), commonly known as a power breaker.

Portable, electrically powered saws must never be carried by their power cable and, when operating, the power cable must always be kept clear of the saw blade. The power plug must be inspected each time, prior to use, for damage and security of the cable. All electrically powered saws should be inspected regularly by a competent electrician.

Care should be taken to ensure that the spring loaded, lower blade guard returns to cover the blade after every cut. This guard, also known as the swivel guard, is designed to uncover the saw teeth when material is being cut, and then automatically cover the saw teeth when the saw is withdrawn. This guard must **never** be wired in the open position.

5.7.7 Chainsaws

Due to their high potential to cause serious injury, the use of chainsaws on building and construction sites is not generally recommended. While they are not illegal, site rules may prohibit their use. Where chainsaws are used on site, they must only be used by operatives who are fully competent, have received specific and nationally approved training, and who are authorised to carry out the intended operation.

A specific risk assessment should be carried out and a written method statement produced for the operation, including the need for full PPE, safe working at heights and, if necessary, a workplace located safely away from other workers.



For specific guidance on the use of chainsaws refer to the HSE leaflet *Chainsaws at work* (INDG317).

5.7.8 Other woodworking equipment



For guidance on the safe use of other specific types of woodworking equipment refer to the HSE woodworking information sheets, which can all be downloaded, without charge, from its website.

5.8 Compressors

Air compressors are used in construction as a power source for a range of tools and equipment. They can be either diesel or electric powered, trailer mounted or static, and range in output from 60 to 900 cubic feet per minute of free air delivered, with working pressures of between seven and 12 bar (100 to 170 psi).

5.8.1 Selection

When selecting a compressor for a particular application the following points should be taken into account.

- How many tools are going to be run from the compressor?
- What is the total airflow required?
- What is the working pressure required?
- Does the compressor need to be trailer mounted for ease of movement?
- Where will the compressor be located?
- Are diesel fumes an issue?
- Are there any noise limitations?
- How long will the compressor be required?

5.8.2 Compressed-air safety

A compressed-air system presents a range of hazards.

- Unintended release of pressure from the air receiver, piping or a loose hose-tool connection.
- Hoses that become disconnected from a tool, whipping in an uncontrolled fashion.
- Noise from compressed air leaks and pneumatic machinery.
- Hand-arm vibration from pneumatic tools (this is also a hazard with electric tools).
- Hot discharge pipes.
- Moving parts of compressor.
- Trip hazard from trailing air hoses.

Some ways to reduce the risks associated with the use of compressors and air tools are shown below.

- Train users in safe working practices, and supervise activities.
- Select low vibration tools from reputable suppliers.
- Ensure guards protecting the belt and pulley drive are in place.
- Provide adequate ventilation.
- Fit a safety valve, pressure gauge and drain valve to air receivers.
- Fit whip checks across all hose connection points.
- Consider replacing quick disconnect hose connections with safer two-stage disconnect components.
- Maintain a written scheme of examination, as required by the Pressure Systems Safety Regulations 2000.
- Wear the correct PPE (the last resort).



Whip checks should be used on every hose joint to prevent break away



When compressed air is misused, it can cause serious injury or even death

A boilermaker was making repairs on a boiler. Upon emerging from the firebox, he disconnected the air hose from a riveting gun and began to blow dust from his clothes. There was one spot on his jacket sleeve that was so embedded in the fabric that the air did not readily remove it. In order to direct more air into the spot, he placed his index finger over the end of the air hose to partially close the opening. Air entered his finger through a small puncture wound, causing swelling and pain. Foreign material, apparently made up of dust and soot from the air, and oil from the compressor, was later removed from the wound. Even with proper medical treatment, gangrene set in and the finger had to be amputated.

As shown by this accident, using compressed air to blow dust or dirt off clothing or the body is a dangerous practice. A strong blast of air can dislodge an eye from its socket, or rupture lungs, intestines or an eardrum. Air forced into the bloodstream can even cause death.



For further guidance on the use, maintenance and examination of compressors and compressed air systems refer to the HSE publications *Safety of pressure systems – Approved Code of Practice (L122)* and *Compressed air safety (HSG39)*.

5.9 Pumps

Pumps are used in the construction industry for a number of tasks (such as dewatering excavations, pumping bentonite for diaphragm walls, pumping concrete and pumping sludge or sewage). Pumps can be diesel, petrol, electric or air powered and range in capacity from around 2 m³/hr to 1,200 m³/hr. They may be trailer or skid mounted and are available in silenced versions and as submersible units.

5.9.1 Selection

When selecting a pump for a particular application the following points should be taken into account.

- What type of fluid is being pumped?
- What is the flow required?
- What are the suction and delivery heads required?
- Does the pump need to be trailer mounted for ease of movement?
- Where will the pump be located?
- Are diesel or petrol exhaust fumes an issue?
- Are there any noise limitations?
- How long will the pump be required?

Providing the pump supplier with this information will assist in getting the correct pump for the application.



For detailed guidance on the safe use of concrete pumps refer to the Construction Plant-hire Association *Code of Practice for the safe use of concrete pumps (BCPG0401)*.

5.10 Lasers

If used correctly, lasers should not pose a health or safety hazard. The operating movement of a rotating laser means that, in practice, it is difficult to look directly at the beam for more than an instant. Static lasers (such as pipe lasers) pose more of a risk. Exclusion zones and warning signs must be in place if high-powered lasers are being used.

Laser equipment is commonly used in the construction industry. It is important to know the different classifications, the dangers and their safe use. The British Standard BS EN 60825-1:2014, *Safety of laser products Part 1: Equipment classification and requirements*, classifies laser products according to the laser beam hazard.

The standard sets out eight classes of laser products, consisting of the following:

- Class 1
- Class 1C
- Class 1M
- Class 2
- Class 2M
- Class 3R
- Class 3B
- Class 4

The classification scheme for lasers indicates the potential risk of adverse health effects; the higher the class number, the greater the radiation hazard posed by the laser.



Rotating laser level. Photo reproduced by kind permission of Hilti

Government guidance states that any Class 1 laser product is regarded as 'safe', for example barcode scanners, laser printers and laser levels. When working with laser equipment, it is important to follow the following points.

- Read operating instructions for the equipment before use.
- Set up laser equipment above or below normal eye level.
- Never point the laser at other persons.
- Keep lasers out of reach of children.
- Never look into a laser beam.

'Hazardous' lasers, which cover all Class 3B and 4 lasers, present a 'reasonably foreseeable' risk of harming the eyes and skin of workers, and are where control measures are needed. These are specified as 'hazardous' because of the potential to cause eye damage including blindness, burns to the skin, and their ability to cause a fire. Workers using this class of equipment and above must be trained and competent.



For detailed guidance on the safe use of laser equipment, refer to the HSE *Guidance to Employers on the Control of Artificial Optical Radiation at Work Regulations (2010)* and examples of 'safe' and 'hazardous' lasers based on definitions in the British Standard BS EN 60825-1:2014.

5.11 Unmanned aerial vehicles (drones)

Small unmanned aerial vehicles (UAVs) are becoming more widely available for construction activity.

More popularly known as *drones*, they can carry out surveys at height and in inaccessible areas. Perhaps most importantly, they can remove the need for people to enter dangerous environments for inspections and surveys.

It must be acknowledged that if drones are not used responsibly they can create dangers for operatives and members of the public. Because of recent incidents, the use of drones is subject to specific safety regulations relating to the way they are operated. These rules are underpinned by UK law.

The regulations are contained within the Air Navigation Order 2016 and there are some specific additional steps that must be taken if a drone is being flown for commercial operations.

In summary, the order requires that anyone carrying out commercial drone flights must have permission from the Civil Aviation Authority (CAA) and abide by the regulations in relation to the size of aircraft being flown, areas of flight, due care and attention for aeroplane flight paths and for the safety of the general public.



A drone in flight

5.11.1 Possible risks associated with drones

Failure of propulsion and/or control could cause the drone to hit people, property or the ground with the potential to cause serious injury or death (for example, a drone incident might cause a fire or a road traffic collision). Improper use of drones poses a threat to, and may breach, the privacy of people and property both on and adjacent to construction sites. This is of greater concern where sites are located in built-up areas that include houses, roads, schools and highways.

Appropriate airspace separation between unmanned and manned aircraft is essential. Regulations govern where, when and how drones may be operated. This includes the height at which drones may be flown, as well as permitted proximities to airfields. Professional and competent drone operators will be well informed as to how to carry out and complete their work safely and in compliance with all current regulations.



Reportable occurrences

Be aware that certain types of incidents and accidents involving a drone may be reportable to the CAA and/or the Air Accidents Investigation Branch (commonly referred to as the AAIB). Your drone operator should know when and to whom such reporting is required. Mandatory reporting is the responsibility of the drone operator.

5.11.2 Reducing the risks associated with the use of drones

The following steps can be taken to reduce the risks posed by operating drones on construction sites.

Undertake basic drone training. Prepare staff who may be responsible for overseeing drone operations on site with a simple and broad appreciation of drone operations and their implications in the construction environment.

Choose a qualified and competent operator. Choose a drone operator (pilot) who is appropriately qualified and competent. Ask to see the operator's qualification and confirm their level of experience of carrying out similar tasks in similar environments.

Check permissions. All commercial drone operators are required by law to be holders of a permission issued by the CAA. Ask to see the operator's permission, check that it is in date and that the operator's planned activities are within the scope of the permission.

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Task feasibility. Ensure that the operator is satisfied that the intended task is feasible for a drone to complete safely, effectively and in compliance with current regulations, and within the operator's permission.

Aircraft suitability and serviceability. Ask the operator to confirm that the drone is suitable for the task, has been regularly inspected and maintained and will function safely in the prevailing site conditions.

Suitable and sufficient resources. Check that the operator has allocated a competent pilot and identified sufficient support resources to complete the aerial data collection safely and effectively.

It is common for a drone operator to be accompanied by support personnel who will help monitor and manage the drone in flight, operate the payload (for example, a camera), and provide assistance on the ground (for example, managing pedestrian and vehicular traffic).



All personnel should be trained and properly briefed by the drone operator before being assigned any duties.

Pre-flight planning. Before allowing any drone flights on site, ensure that your operator has undertaken and completed preflight planning and an on-site survey. Such planning should include a detailed risk assessment and the proposed method for carrying out the aerial work.



All operational procedures should be detailed within the drone operator's operations manual, sight of which is recommended, prior to allowing any flights to be conducted.

Site induction. See that the operator and supporting personnel receive a full site induction before being allowed to carry out any work. Make sure that the operator is made aware of all site hazards which could pose a risk to the health, safety and welfare of personnel and/or which could compromise flight safety. Consider overhead hazards (for example, cranes, overhead power lines, tall structures and tree canopies).

Minimise flight time. Undertake detailed planning with the drone operator to minimise flight time and support the safe completion of the task.



Reduced flight time generally equals reduced flight risk.

Establish a safe operating zone for the operator and support personnel, liaising on safe on-site locations for the following.

- An area for the operator to store and prepare equipment.
- A primary take off and landing zone.
- A secondary take off and landing zone (in case the primary location becomes unavailable).
- An appropriate place for the pilot and other personnel to stand, allowing safe and effective work whilst avoiding conflict with other site activities.



A safe, clearly marked take-off and landing zone



Avoid unnecessary communication with the pilot during flights. This will allow them to focus on safe operation, minimising flight safety risks.

5.11.2.1 Minimise risk to those on the ground

Flying directly over people on the ground should be avoided. If flying above people cannot be avoided, it is essential that appropriate safety measures are put in place or that the persons affected are placed under the control of the person in charge of the drone.



Risk mitigation measures to reduce the likelihood of causing harm include choosing a time when there are fewer people in the operating location (for example, during lunch or break periods, or at the start or end of the day).

5.11.3 On-site communications and emergency procedures

Discuss, agree and establish robust communication between the drone operator and the site management team. These must be appropriate to the site environment, the task and the operating conditions and risks. Discuss with the drone operator what emergency procedures are in place, in the event of an incident or accident involving the drone. Make sure that you are satisfied that these are appropriate, and that everyone understands what they need to do if there is an emergency.

5.11.4 Pilot considerations

Pilot health. Check that the pilot is satisfied that they are in a fit state of health to be able to operate the drone safely.

Avoid coercion or pressure on the pilot. The drone pilot must be satisfied that any proposed flight can be safely made. The pilot must not be pressurised or coerced to contravene safe operating protocols or to operate an aircraft when they do not consider it safe to do so.



For further information about drones and UAVs, visit the CAA website and the Air Accidents Investigation Branch website.

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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

A significant number of fatalities occur every year as a result of people being struck by moving vehicles. Many of these fatalities are caused by mobile plant. The majority could have been avoided by effective planning and control of mobile work equipment, adequate checking, inspection and maintenance of the plant, and thorough training, assessment and briefing of personnel.

This chapter gives an overview of the legal requirements for mobile work equipment and the steps required to ensure that these obligations are met so that mobile work equipment can be used safely and efficiently.

6.1 Introduction

The use and movement of mobile work equipment, whether over site roads or within specific areas, involves different and continually changing hazards. These have the potential to cause harm, such as persons being struck or run over, or thrown from or crushed by moving plant or equipment.



Problems with mobile work equipment often occur because of poor communications when planning the work and briefing operators.

6.2 Important points

- Many accidents occur with mobile work equipment because operations were not properly planned in advance.
- Mobile work equipment should only be operated by trained, competent and authorised persons who have been properly briefed on the task to be carried out.
- The rated capacity or safe working load (SWL) of any item of mobile plant must never be exceeded.
- Mobile work equipment designed for specific operations must only be used for those operations. The design limitations and permitted modes of operation specified by the manufacturer must be complied with.
- All mobile work equipment must be subjected to a schedule of maintenance, inspections and thorough examination.
- Attention must also be paid to site features, such as overhead cables, excavations, unstable ground conditions and traffic and pedestrian routes.
- Mobile work equipment, first supplied after 1 January 1995, must be CE marked and meet the essential health and safety requirements of the Supply of Machinery (Safety) Regulations 2008. Any equipment purchased from 1st January 2021 within the UK must show the UKCA marking. If imported from the EU, the product must also show the UKCA mark alongside the CE mark. Consequently, most items of lifting equipment are designed to a designated standard (such as BS EN 474-5:2022 for excavators, BS EN 1459 for telehandlers and BS EN 280-1:2022 for mobile elevating work platforms (MEWPs)).

6.3 Legislative requirements



For a brief summary of the legislative requirements for mobile work equipment refer to Chapter C05 Work equipment and hand-held tools.

For more detailed information on legislation refer to Section A: *Legal and management*.

6.4 Types of mobile plant

By definition, mobile plant is plant that moves around a construction site, generally under its own power, and is driven or controlled by an operator. Mobile plant is subdivided into the four areas listed below.

6.4.1 Ride-on plant

Ride-on plant is where the operator sits/stands on the machine or in a cab (examples being rollers, excavators, loading shovels, rough terrain cranes, crawler cranes, self-propelled MEWPs, telehandlers, road planers, asphalt spreaders and dumpers).

Plant that is not designed to carry passengers should be provided with a notice clearly stating 'No passengers'.

People can only be carried if the vehicle has been designed for such a purpose.

Passengers and drivers must not be carried unless proper seating, correctly fitted to the main structure of the vehicle, is provided.

Where passengers are permitted to be carried, additional restraining devices must be provided for each person.

Where on-board work has to be carried out, seating should be provided, if possible, with work platforms fitted with suitable barriers or guard-rails to stop operatives falling from the equipment whilst it is in motion.

6.4.2 Pedestrian-controlled plant

This type of equipment relies heavily on the experience and competence of the operative for its safe operation. Small pedestrian-operated rollers, vibrating plate compactors and surface grinders are among the types of equipment within this category.

Common accidents involving pedestrian-operated equipment occur when operators are crushed between the machine and adjacent obstructions. This is one reason why the regulations require efficient emergency stop controls to be fitted.

It is also essential that operators are provided with, and wear, strong protective footwear to prevent their feet from being injured.



This micro-surfacing machine is designed to carry more than one person

6.4.3 Remote-controlled plant

The use of remote-controlled equipment is undertaken by specialists and is increasingly common on construction sites. It removes the operator from areas of danger as the equipment can be manoeuvred into hazardous or aggressive environments (such as enclosed spaces or unstable structures). The operator can then position themselves away from the dangers of falling material, dust, noise, fumes, vibration and moving parts of the machine. The equipment takes all of the risk and the operator is protected against hearing damage, deafness, hand-arm vibration syndrome, whole-body vibration and crushing from falling structures.

Common activities using remote-controlled equipment include demolition, core drilling, floor sawing and excavations in hazardous locations (for example, where there is a risk of rock fall).

6.4.4 Lifting equipment



For further information on the lifting aspects of mobile plant that is also lifting equipment (such as mobile cranes, lorry loaders, MEWPs, telehandlers and excavators used for lifting) refer to Chapter C07 Lifting operations and Chapter C08 Lifting equipment.

6.5 Planning and use of mobile work equipment

The **Management of Health and Safety at Work Regulations 1999** require that all tasks carried out at work are subject to a risk assessment to identify hazards and assess the risks associated with the task.

Planning a safe site should begin before the construction phase. In accordance with the **Construction (Design and Management) Regulations 2015** there are specific duties placed on all of those involved within the construction process, including clients, principal designers, designers, principal contractors and contractors. The planning process for the use of mobile plant is similar to that for most other tasks, and consists of the following steps.

Step 1	Identifying the task to be undertaken.
Step 2	Identifying the hazards associated with the task.
Step 3	Carrying out a risk assessment.
Step 4	Identifying and implementing control measures.
Step 5	Developing the method to be used.
Step 6	Recording the planning in a method statement (including any contingency plans for rescue).
Step 7	Communicating the plan to all persons involved.
Step 8	Reviewing the plan before the task starts, periodically and incorporating any changing circumstances.

6.5.1 Practical measures for safe workplaces

The following practical measures should be considered to ensure that mobile work equipment can be used safely on site.

- Pedestrian routes should be established on site to facilitate safe pedestrian movement and access to work areas.
- Pedestrian routes should be segregated from mobile plant and vehicles, either by a safe distance or by physical barriers.
- Traffic routes should be planned in order to minimise disruption to neighbours, protect the public, avoid congestion and risk of collision. These routes should be kept free of obstructions and properly maintained, with access points restricted and clearly marked.

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- Appropriate speed limits should be introduced.
- Set up vehicle-only areas, especially where space is limited or traffic is heavy.
- One-way traffic systems should be implemented, where possible.
- Parking places should be designated for delivery vehicles and other vehicles left temporarily on site.
- The operating area should be clear, as far as possible, on all sides.
- Operators must be informed of the location of any overhead power lines, sensitive areas, underground cables, sewers, ducts or services before operations commence.
- Excavations should be fenced or otherwise guarded.
- Ground conditions should be stable and sufficiently level for the operations being carried out and the equipment being used.
- Where equipment fitted with outriggers or stabilisers is used, the load-bearing capacity of the ground should be assessed so that the imposed loads do not exceed this capacity. An additional safety factor should also be applied.
- Where site vehicles are employed in tipping material into excavations, baulks of timber or other effective blocks (stop blocks) should be provided to prevent the vehicle over-running the edge.
- Excavations may require extra support or shoring to prevent the weight of adjacent vehicles causing a collapse (surcharging).



For detailed guidance on the measures to be taken in planning for the use of mobile work equipment refer to the Health and Safety Executive (HSE) guidance *The safe use of vehicles on construction sites (HSG144)* and *Protecting the public: your next move (HSG151)*.

6.5.2 Selection of mobile work equipment

In selecting the work equipment, every employer must take account of the working conditions and the risks to the health and safety of persons who are working in the vicinity of where the mobile equipment is to be used.

The design of some vehicles presents hazards, such as restricted visibility, lack of driver protection, exposure to noise and vibration and the risk of overturning.

6.5.2.1 Choosing the right equipment for the job

This is an essential part of effective management for mobile work equipment. The equipment selected needs to be capable of performing its designated tasks safely. Some important factors to consider are given below.

- Stability under all foreseeable operating conditions (for example, ground and weather conditions).
- Safe access to and from the cab and other working locations on the vehicle.
- Effective braking systems.
- Adequate visibility for the driver all around the vehicle, especially when reversing.
- Headlights, a horn, windscreen wipers and warning devices (such as reversing alarms).
- Physical guards to protect dangerous parts (such as power take-off shafts, chain drives, trapping points and exposed exhaust pipes).
- Protection for the driver from work hazards (such as working at height and falling from the vehicle, falling objects and the effects of the vehicle overturning).
- Protection for the driver from the weather, noise, vibration, fumes and dust.

Manufacturer's specifications need to be considered when selecting vehicles for construction and civil engineering work. In particular, load and stability limits need to be taken into account for vehicles intended to be used on uneven and sloping ground. Some vehicles, especially those involved in lifting operations (such as some lift trucks and telescopic handlers), require flat, compacted surfaces to operate safely.

6.5.2.2 Concrete pumps

All concrete pumping pours should be planned to ensure that they are completed safely and that all foreseeable risks have been taken into account. Planning should be carried out by personnel who have the appropriate expertise. In cases of repetitive concrete pours, this planning may only be necessary in the first instance, with periodic reviews to ensure that no factors have changed.



For detailed guidance on the safe use of concrete pumps refer to the Construction Plant-hire Association's *Code of Practice for the safe use of concrete pumps*.



Visit the Concrete Structures Group website for more information and guidance on the dangers of concrete pumping and pours, checks and inspections of pipelines, clearing of blockages, and the cleaning out process.

6.5.3 Braking systems

Effective braking systems are essential for the safe use of vehicles. In accordance with the Road Traffic Act 2018, parking brakes should be fitted on trailers over 0.75 tonnes maximum gross capacity. Where parking brakes are not fitted, trailer wheels need to be chained or locked to prevent movement when the trailer is parked. Wheel chocks should be used to prevent unintended vehicle and trailer movements. Trailers with maximum gross weights between 0.75 tonnes and 3.5 tonnes should have at least an overrun brake (such as an inertia brake), whereas trailers over 3.5 tonnes should be fitted with braking systems linked to the towing unit.

6.5.4 Roll-over protective structures

Where there is the risk of mobile work equipment rolling over, it should be fitted with roll-over protective structures (ROPS), such as reinforced cabs or a roll-cage, thereby minimising the risk of a person being crushed. The structure should give adequate clearance for operatives to escape if the equipment does completely overturn.

Where there is a risk of an operator being thrown then crushed by equipment rolling over, a suitable restraining system (such as a seat belt) should be fitted **and worn by the operator**. New mobile plant is fitted with green flashing lights on the outside of the cab that indicate that the seat belt is being worn.

This requirement may also apply to equipment fitted with a fully enclosed cab if there is a risk of a person who is thrown from their seat being injured by coming into contact with the internal cab structure.

Restraining systems must be fixed to a designated anchorage point on the main structure of the equipment.

In areas of limited access, ROPS may be removed only if a suitable and sufficient risk assessment has been carried out and there is no risk of overturning. The ROPS structure should be refitted when the machine is put back to work in areas where space is not restricted. An assessment should be made of the likelihood and potential consequences of a vehicle roll-over or overturn on site, and the nature of the mobile work equipment and any attachments or accessories fitted to it.



ROPS and FOPS cab

e.g. Wearing seat belts

Telehandler cabs are provided with seat belts to restrain the operator and reduce the risk of serious injury in the event of the machine overturning. A survey of telehandler accidents in the UK carried out by the Health and Safety Executive (HSE) shows that in seven years there were at least 72 lateral overturns of telehandlers. In three cases, the operator was killed by being ejected from the cab and crushed; in a further two cases, the operator was ejected from the cab and received serious crushing injuries. Had these operators been wearing their seat belts, their deaths or injuries would probably have been avoided.

6.5.5 Falling object protective structures

Where there is the risk to persons operating mobile work equipment of being struck by falling material, falling object protective structures (FOPS) or a manufacturer's strengthened cab must be fitted to stop any falling material striking the operator.

6.5.6 Prevention of unauthorised start-up

Unauthorised persons must be prevented from starting up self-propelled mobile work equipment wherever it is parked. Additionally, it must be possible to isolate the drive mechanism. This can be achieved by simple measures (such as the removal of ignition keys or starting handles), the use of more sophisticated systems (such as keypad or smart card isolators), or remotely controlled systems and effective vehicle immobilisation systems and key custody procedures.



Some old designs of site dumpers allowed the vehicle to be knocked easily into gear as the driver dismounted.

6.5.7 Lighting and warnings

When self-propelled mobile work equipment is used in the dark or in reduced visibility, lights must be fitted and used to enable the work area to be adequately illuminated. In addition, amber flashing beacons that give warning of the presence of the vehicle should be fitted and used. The site traffic management plan may dictate that construction vehicles are fitted with audible reversing warning devices or closed-circuit television (CCTV) cameras.

MOBILE WORK EQUIPMENT

6.5.8 All-round visibility

Accidents involving mobile work equipment and pedestrians frequently involve poor visibility where the plant operator has not seen the pedestrian. Current standards aim to ensure that the operator of the plant or vehicle has all-round vision from the operating position.

Therefore, plant needs to be assessed to ensure that the existing levels of visibility meet the requirements of the Provision and Use of Work Equipment Regulations 1998, published guidance and your own safe system of work.



It is generally accepted that the operator should be able to see, at all times, an object positioned 1 m above ground level and 1 m away from the plant through 360° visibility.



All-round visibility

All-round visibility can usually be achieved by the use of aids, such as additional mirrors, convex mirrors or CCTV.

Where this is not possible, consideration should be given to providing a competent, qualified vehicle marshaller working exclusively with the plant operator and implementing exclusion zones to prevent unauthorised access into any danger zones (*for further information on plant safe zones refer to 6.6*).

6.5.9 Forward-tipping dumpers

Amongst mobile work equipment, forward-tipping site dumpers are worthy of special note because they are involved in a disproportionately high number of accidents.

This applies to both rigid-frame and articulated dumpers.

In recognition of the high number of accidents, and to improve the welfare and working conditions of operators, dumpers are now available with protective cabs and enhanced safety features.



Over 60% of dumper-related deaths involve the driver when the vehicle overturns. Most other deaths occur when pedestrians are struck by the dumper whilst it is in operation.



Forward-tipping dumper - bad practice: travelling with a raised skip

The common causes of these accidents are shown below.

- Overturning on slopes, rough ground and at the edges of excavations.
- Travelling with a high-lift skip in the raised position.
- The operator failing to observe pedestrians, who are then run over by the front wheels.
- The operator being thrown from the vehicle whilst travelling over rough or sloping ground.
- Operator error due to lack of experience and training, including accidental operation of the controls.

Efforts to reduce dumper accidents should focus on the following.

- Selecting the correct machine for the job.
- Effective site management and control of how dumpers are operated generally.
- Ensuring that machines are routinely inspected, serviced and withdrawn from use if unsafe.
- Ensuring that site roads and operating areas are suitable, maintained and safe for the use of dumpers.
- Ensuring that only trained, competent and authorised operators are allowed to operate dumpers.



For detailed guidance on the safe use of forward-tipping dump trucks refer to the HSE publication *Construction site transport safety: safe use of site dumpers (CIS52)* and the Construction Plant-hire Association's *Staying safe when operating forward tipping dumpers*.

The HSE has produced an Approved Code of Practice (ACoP) on *Rider-operated lift trucks (L117)*, which outlines the main legal requirements of use.

6.5.10 Quick hitches

The quick hitches fitted to excavators and other earthmoving equipment make a valuable contribution to the construction process by enabling buckets and other attachments to be attached and detached rapidly.

Unfortunately there have been a significant number of accidents involving the use of quick hitches, which have included fatalities.

When using quick hitches, the important safety precautions (listed below) should be observed. These precautions are applicable to all types of quick hitch – manual, semi-automatic and all fully automatic types, including those that lock onto both pins.

All types of quick hitches can be safely used by competent people within a safe system of work. This will involve planning, training, instruction, communication, supervision and maintenance.



Site management should familiarise themselves with the different types of quick hitches, and any related company policy, to assess and monitor what contractors should be doing to manage the risk.

- Keep other workers away from the machine's working range. Never work under the bucket or attachment and ensure that the operator isolates the controls if there are other workers in the machine's working range. Ensure that your site supervisors or vehicle marshalls enforce this rule.
- Make sure that your operators are aware of their personal and legal responsibility to use all quick hitches safely. Serious misuse of quick hitches should always be treated as gross misconduct, which could lead to dismissal or individual prosecution.
- Prohibit practices (such as pick and place) that involve moving attachments without properly engaging the locking devices.
- Check that all operators confirm that the quick hitch is in place every time they change an attachment. It is essential that this is done by close inspection at the quick hitch, which will normally entail the operator getting down from the cab. It is good practice to make this a requirement for all quick hitches.
- Check that site supervisors understand that the operator has to fully confirm that the quick hitch is in place and allow operators time to do this.
- Check that site supervisors monitor that operators are confirming proper engagement, by spot checks. Avoid conflict by ensuring that all operators understand and buy into the reasons why the supervisor has to carry out spot checks.
- Check that machines are in good order, and all quick hitches are included in the maintenance plan and subject to proper maintenance.
- Check that operators and maintenance personnel have specific manufacturer instructions for the combination of machine, quick hitch and attachment.
- Confirm that operators are competent to use the specific quick hitch in accordance with the manufacturer's instructions and the safe system of work.



Keep other workers away whilst changing buckets or attachments



e.g. Using quick hitches with other attachments

Contractors may want to use quick hitches with attachments other than a bucket in normal configuration.

However, if a specific application (such as the fitting of vibratory hammers or reversing of buckets) is not explicitly referenced by the manufacturer, the user should assume that it is not allowed, and should not use the quick hitch for this attachment or application.

Quick hitches should be used only with attachments and in applications as specified by the manufacturer.



It is essential that operators get out of the cab to physically ensure that all quick hitches are securely locked before starting work with a newly fitted attachment.

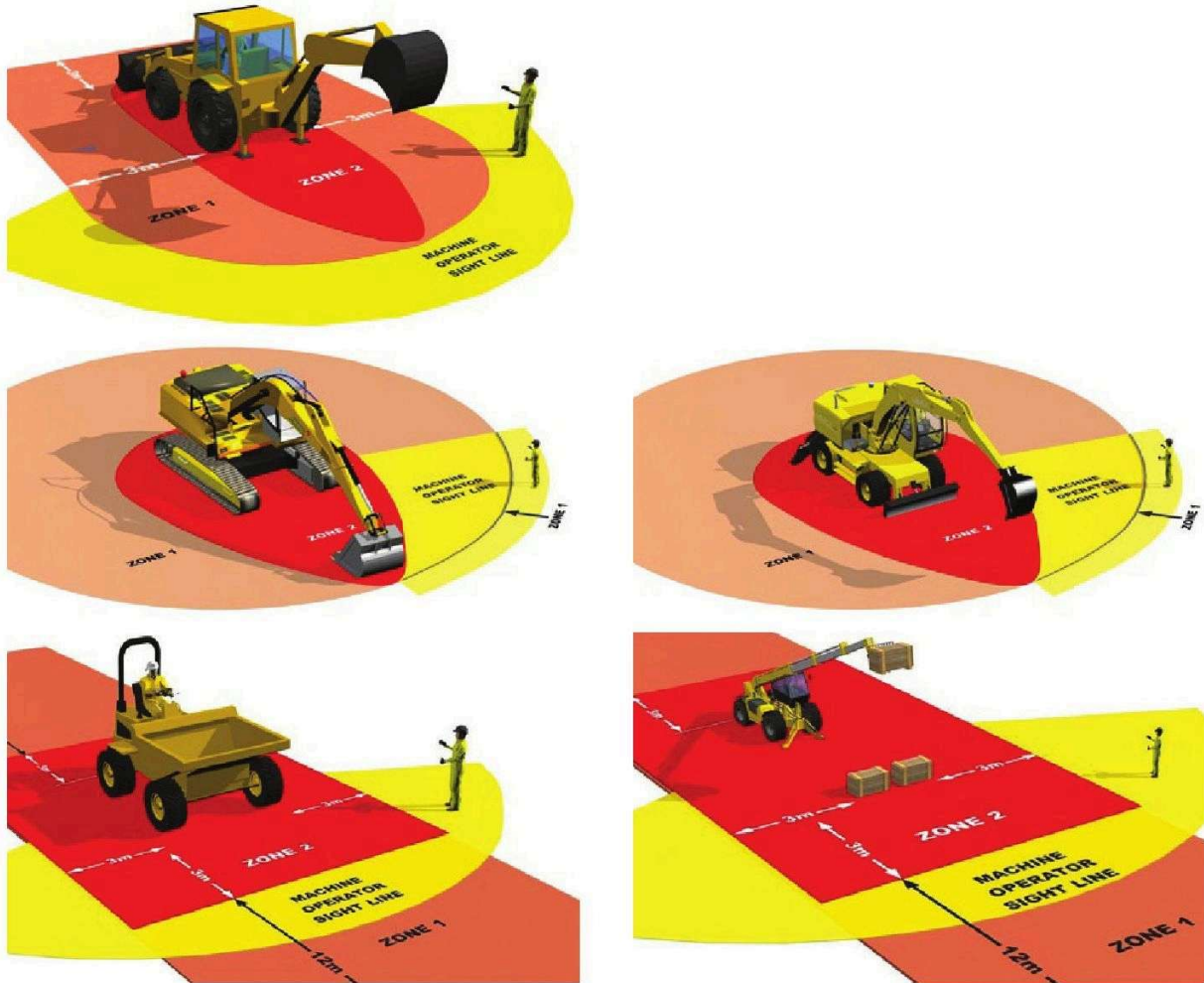


For detailed guidance on quick hitches refer to the Strategic Forum for Construction's *Safe use of quick hitches on excavators* (Best practice guide).

MOBILE WORK EQUIPMENT

6.6 Plant safe zones

The following diagrams are for guidance purposes, and show the safe zones applicable to a range of plant machinery likely to be used on site.



(Reproduced from The Delivery Hub health, safety and environment – Raising the bar 1 – Plant and equipment from the Highways Agency, under licence of the Controller of Her Majesty's Stationery Office.)



Always signal the operator and receive a positive response before entering Zone 1.

Keep out of Zone 2 at all times.

6.7 Training and competence of personnel

It is essential that all involved in planning, supervising and carrying out mobile plant operations are adequately trained and competent. There are a number of nationally-recognised card schemes that provide evidence of training. The Construction Plant Competence Scheme (CPCS) also provides proof of competence, evidenced by endorsement with the Construction Skills Certification Scheme (CSCS) logo on their card. It is, however, each employer's responsibility to ensure that employees and those working under their control are competent to undertake the tasks they have been allocated.

It is also important that mobile plant operators hold appropriate qualifications and endorsement for each different plant category they operate. Before allowing employees to operate plant, check that they are trained and that their industry card is endorsed with the right category (for example, 360 excavator, dumper, or telehandler). In addition, operators must be familiar with the specific machine they are operating.

Mobile work equipment comes in many shapes and sizes, with significant differences in control layouts and operating characteristics. It is therefore essential that operators are given **adequate familiarisation** before use. This is generally carried out by an experienced person employed by the equipment owner. Training should include control layouts for particular machine models, starting and stopping (including emergency stops), operating characteristics, emergency procedures (such as for spills, fire and overturn), and inspection and maintenance checks. Records should be maintained for all familiarisation training, clearly indicating makes and models covered.



Details of the Construction Plant Competence Scheme (CPCS) can be found online.



Worker's head struck with an excavator bucket

A construction worker has been prosecuted after he struck another worker on the head with the bucket of an excavator. He was prosecuted after an investigation found he had been using a mobile phone whilst operating the excavator on a building site, and had not noticed his colleague.

The construction worker was working alongside a colleague who was driving a site dumper truck. The operation involved excavating and dumping material into the truck for transport to another location at the site. The court heard that the driver of the dumper truck had returned to the excavation site to await the next load of material. The construction worker had been using his mobile phone and, not realising his colleague had returned, rotated the upper body of the excavator, causing the metal bucket to strike the driver on the side of his head.

The driver sustained multiple fractures to his jaw, as well as a punctured and collapsed lung. He was hospitalised for 10 days and did not return to work until 14 months later, with further surgery still required on his jaw. The construction worker was ordered to pay compensation of £2,500 to the injured worker, and costs of £1,554 after pleading guilty to a single breach of the Health and Safety at Work etc. Act 1974.

Speaking after the hearing, the HSE inspector said: 'Construction site vehicles are extremely powerful and, if the operator becomes distracted, can be highly dangerous. Road users are rightly banned from using mobile phones when driving cars. It's clearly important that those in control of machinery – weighing up to 40 tonnes in some cases – need to be equally attentive and concentrate solely on the job at hand. This incident could easily have been avoided if the operator had followed site rules and not become complacent about his responsibilities when operating his vehicle.' (Source: HSE.)



For detailed guidance on plant operator competence refer to the Strategic Forum for Construction Plant Safety Group's *Competence to operate construction plant* (Good practice guide).

6.7.1 Medical fitness

People who operate mobile equipment have the potential to injure both themselves and others in the workplace and the surrounding area. Ensuring that operators are medically fit to operate is primarily an issue for employers, who have a duty to ensure that any employee is physically and mentally capable of undertaking the tasks they are required to carry out. Others in the construction sector (such as principal and sub-contractors) will also have an interest in ensuring operators have an appropriate level of fitness.



For detailed guidance on medical fitness refer to the Strategic Forum for Construction Plant Safety Group's *Medical fitness to operate construction plant* (Good practice guide).



Operators may need to participate in an assessment to ascertain medical fitness

6.7.2 Supervision

An essential part of the safe use of mobile work equipment is supervision of the operator. The supervisor is required to re-brief the operator and others on the task and ensure they are put to work safely.

- Reinforce the main elements of the safe system of work, including:
 - strict adherence to exclusion zones
 - the correct safe methods when fitting/removing attachments (such as the use of quick hitches)
 - consulting the operator and others regarding any issues/comments they have in adhering to or the effectiveness of the safe system of work and, where appropriate, instigate changes.
- Check that the operator has undertaken the relevant daily checks.
- Check the required maintenance has been undertaken.
- Ensure the inspection/maintenance log/check sheet has been completed and signed. The supervisor's signature on the check list/log may be used as verification that they have carried out the check.

As part of the supervisor's ongoing duties, they must, throughout the day, monitor that the safe system of work is being adhered to. This should include maintenance of exclusion zones, checking that no-one is working below attachments at any time, and ensuring quick hitches are being used according to the manufacturer's recommendations, including correct use of any safety pins or locking devices. They should regularly check that the manufacturer's manual for operating the machine is in the cab, that any relevant instructional decals (durable stickers) are displayed in the cab and/or on the machine, and that the next service or inspection date has not expired. The daily checklist for the machine will provide evidence of some of these points.

6.8 Maintenance, checks and inspections

Maintenance, checks and inspections of mobile work equipment are essential if they are not to deteriorate, break down and ultimately fail.

All maintenance should be carried out by a competent person who is familiar with the equipment and has the knowledge and experience to identify and repair existing or potential faults.

The maintenance should be in accordance with the manufacturer's instructions and be carried out, as a minimum, at the recommended intervals.

All mobile work equipment should have a pre-use check carried out daily or at the start of each shift and any defects reported to a supervisor. This is normally undertaken by the operator who should be trained to carry out the checks.

In addition to pre-use checks, inspections should be carried out at appropriate intervals, either as part of the maintenance process or at more frequent intervals.

The frequency of inspections will depend on the working environment and assessment of the risk of failure of the equipment, and should be guided by the manufacturer's instructions.

The results of all checks and inspections should be recorded and there should be a system in place to ensure that all reported defects are rectified and recorded.

Keep the following points in mind when planning for checks, maintenance and inspections.

- The equipment's potential to cause serious harm (powered machinery, such as a circular saw, will require more regular attention than a hammer).
- Type, class and complexity of equipment (some equipment requires weekly maintenance, others less frequent maintenance).
- Equipment in regular use will probably require more frequent maintenance than equipment used infrequently.
- Likely deterioration of work equipment when not in use.
- The environment in which the equipment is used (for example, an excavator used in a civil engineering project, such as sea defence work during the winter, will have the potential to deteriorate faster than an identical machine used in a less hostile environment).



Each item of mobile work equipment should have a history file containing all records of maintenance, inspection and thorough examination, so that maintenance trends can be established and adequate maintenance demonstrated in the event of an accident.



For guidance on construction plant handover and inspection reports refer to Appendix A.



A mobile lighting tower may not have a designated operator but should still be subject to pre-use checks

Appendix A – Construction plant handover and inspection report guidance

These guidance notes are provided to assist the persons appointed by the site manager to complete reports.

Mobile plant is defined as mechanical or electrical plant that is self-propelled. The types of plant typically covered by the mobile plant handover report are as follows.

- All types of excavator.
- All forward or rearward tipping dumpers/trucks.
- All types of cranes.
- All types of compacting rollers.
- Rough terrain forklift trucks.
- Tractors and their trailer units.

In the case of plant provided by a plant hirer, then, at the site manager's discretion, a competent person appointed by the plant hirer may conduct the inspection and complete the report.

In the case of plant owned or hired by a contractor, then a competent person employed by the contractor may conduct the inspection and complete the report.

All completed mobile plant handover reports should be retained for inspection on site.

The report is produced in a checklist format for ease of use and is intended to provide a means by which the condition of all plant delivered to site can be checked by the appointed person. All questions should record an answer, with a suitable comment or action included when necessary.

The appointed person is also required to confirm the validity of relevant statutory documentation (such as test certificates, thorough examination and inspection reports).

An aide-memoire is provided below to assist the appointed person on statutory requirements that apply to certain items of plant.

Plant	Thorough examination	Comments	Documentation
All types of cranes used for lifting people and MEWPs.	Current six-monthly thorough examination report.	■ Fitted with suitable devices to prevent persons being crushed, trapped, struck or falling when riding or working from carrier. ■ Provision of suitable devices to prevent risk of a carrier failing. ■ Safe means of escape for persons trapped in a carrier.	Evidence of daily or weekly inspection and six-monthly thorough examination, including certification of the rated capacity (SWL).
Excavators used as cranes without any restriction and a fixed or variable rated capacity (SWL) greater than one tonne.	Current 12-monthly thorough examination report.	■ Fitted with controlled lowering devices (check valves). ■ Provided with a properly fitted hook or lifting shackle marked with the rated capacity (SWL). ■ Supplied with a register for the weekly inspections of the excavator and, where fitted, rated capacity indicator. ■ Have the rated capacity (SWL) clearly marked on the machine or displayed in the cab.	Evidence of daily or weekly inspection and 12-monthly thorough examination, including certification of the rated capacity (SWL).
All types of cranes and excavators used as cranes without restrictions and a fixed rated capacity (SWL) of one tonne or less.	Current 12-monthly thorough examination report.	■ Provided with properly fitted hook or lifting shackle marked with the rated capacity (SWL). ■ Supplied with a register for weekly inspections of the machine and, where fitted, a rated capacity indicator. ■ Have the rated capacity (SWL) clearly marked on the machine or displayed in the cab.	Evidence of daily or weekly inspection and 12-monthly thorough examination, including certification of the rated capacity (SWL).



Following completion of the report at receipt of the plant on site, formal inspection/examination must be repeated at least at the intervals specified by legislation or policy.



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

There are a number of fatalities and a significant number of specified injuries in the construction industry each year as a result of poorly-planned lifting operations. In addition, many more incidents lead to property, material and equipment damage, resulting in costly delays.

The majority of these could have been avoided by effective planning and control of the lifting operations, thorough examination of lifting equipment and thorough training, assessment and briefing of personnel.

This chapter gives an overview of the legal requirements for lifting operations and the steps required to ensure that these obligations are met so that lifting operations can be carried out safely and efficiently.

7.1 Introduction

Experience has shown that lifting operations can be a hazardous activity if not properly planned and carried out.

Safe lifting operations will depend upon the following.

- Thorough pre-planning of each lifting operation.
- The availability of suitable lifting equipment, which is properly maintained.
- The provision of adequate information, instruction, training and supervision for everyone involved.
- Compliance with safe systems of work, as detailed in risk assessments and method statements (lifting plan).

Although regulations require that safe systems of work are developed, accidents occurring during lifting operations indicate that all too often these are not in place or are not complied with.

Lifting operations have the potential to cause death and serious injury to site workers and members of the public off site. In addition to the terrible cost in human suffering, accidents have a financial cost.

There is a strong business case for improving safety performance in this area.



Problems with lifting operations often occur because of poor communication during planning and when lifting.



The requirements for installing, operating, checking, maintaining, inspecting and thoroughly examining specific types of lifting equipment and lifting accessories are covered in Chapter C08 Lifting equipment.

7.2 Important points

- Many accidents occur during lifting operations that were not properly planned in advance.
- All site features (such as overhead cables, unstable ground conditions and adjacent properties) must be identified and adequately controlled as part of the planning process.
- All lifting operations must be carried out by, and under the control of, trained and competent persons who have been properly briefed on the lifting plan.
- Effective communication during any lifting operation is vital to ensure the safety of those involved and to protect people in the vicinity. This also applies to the planning phase.
- Specific requirements apply to the lifting of people.
- A contract lift will transfer the majority of the legal responsibility for carrying out the lifting operation in a safe manner to the contract lift company. Companies without the specialist knowledge to plan lifts safely should always procure contract lifts.



The collapse, overturning or failure of any load-bearing part of a hoist, crane or other lifting equipment is a reportable dangerous occurrence, even if nobody is injured.



For further information refer to Chapter A13 Accident reporting and investigation.

7.3 Legislative requirements

A brief summary of the main legislative requirements for lifting operations is listed below.

7.3.1 Provision and Use of Work Equipment Regulations 1998 (PUWER)

PUWER are concerned with such matters as the selection of suitable work equipment, safeguarding of dangerous parts of machinery, provision of appropriate controls, training of operators, and the inspection and maintenance of work equipment, including lifting equipment and accessories.



For further guidance on PUWER refer to the Health and Safety Executive's (HSE's) ACoP and guidance *Safe use of work equipment* (L22).

7.3.2 Lifting Operations and Lifting Equipment Regulations 1998 (LOLER)

LOLER deal with the specific risks arising from the use of lifting equipment (including lifting accessories). They build upon PUWER in many areas and apply to all lifting operations, equipment and accessories. LOLER requirements are shown below.

- All lifting operations are planned, supervised and carried out safely.
- Pre-use checks and inspections at appropriate intervals are carried out on lifting equipment.
- Specific requirements are applied to the lifting of people.
- Lifting equipment must be thoroughly examined by a competent person (equipment examination) on specified occasions and at specified intervals.



For further guidance on LOLER refer to the HSE's ACoP and guidance *Safe use of lifting equipment* (L113).



For a list of definitions refer to Appendix A.

7.4 Planning of lifting operations

The siting, setting up and use of equipment for lifting operations requires careful planning to ensure the safety of all involved and others who could be affected. One person with sufficient training, practical and theoretical knowledge and experience should be appointed to be responsible for planning and supervising the tasks, known as the *competent person* in LOLER or the *appointed person* in the *Code of Practice for the safe use of cranes* (BS 7121). For the purposes of clarity in this chapter, *competent person* and *appointed person* are referred to as the *competent person (lift planner)*. Where reference is made to a competent person in relation to equipment examination, they will be referred to as *competent person (equipment examination)*. The competent person (lift planner) must ensure that the planning for each task includes the following.

- Identifying the task to be undertaken.
- Surveying the site.
- Identifying the hazards associated with the task.
- Categorising the lift (1, 2 or 3, dependent on complexity).
- Carrying out a risk assessment.
- Identifying the need for over-sailing licences for neighbouring properties (see 8.5.1.1 for more detail).
- Identifying control measures.
- Developing the method to be used.
- Selecting lifting equipment and lifting accessories.
- Recording the lifting plan with a method statement (including any contingency work for rescue).
- Communicating the plan to all persons involved.
- Reviewing the plan immediately before the task starts and responding to any changing circumstances.



Lifting operations, no matter how large or small, must be properly planned

The planning of lifting operations includes the consideration of 'temporary works', such as working platforms, outrigger pads, lifting beams and load spreaders (see *Chapter C04 Temporary works*). This necessitates liaison between the appointed person (competent person/lift planner) and the TWC(s).

LIFTING OPERATIONS

7.4.1 Identifying the task to be undertaken

As the first stage in the planning process, the task to be undertaken should be clearly identified, together with the location and sequence.

Certain lifting operations require particular care and attention. Further advice can be found in BS 7121 Part 1. The competent person (lift planner) should be familiar with these and plan the lift accordingly.

7.4.2 Surveying the site

The planning of a lift may involve a site survey, carried out by the competent person (lift planner) or their representative. This involves visiting the location where the lift is to be carried out so that both the task and any hazards involved can be identified. For simple lifts the remainder of the planning process may be completed at the same time, whilst for more complex jobs the person carrying out the survey may need to complete the process off site. The survey should include the assessment and confirmation of ground conditions (ground bearing capacity).

A further site survey may be required immediately before the lifting operation is to be started to ensure no changes have been made to the environment.

7.4.3 Identifying the hazards associated with the task

The hazards associated with the task should be identified. These might be associated with the location where the work is to be carried out, the nature of the lifting equipment, the load to be lifted or the people associated with the task or located in the vicinity.

7.4.4 Categorising the lift

To enable lifts to be planned, supervised and carried out effectively, three categories of lift are detailed below. The category into which a particular lift will fall depends on the assessment of the hazards associated with both the environment in which the lift is to be carried out and with the load.

As shown in the table, increases in environmental and/or load complexity (the *complexity index*) will lead to the lift being allocated a higher category. Having identified the hazards associated with a particular lift, a hierarchy of control measures should be applied to eliminate or control those hazards.

7.4.4.1 Basic lift

For a basic lift, the duties of the competent person (lift planner) should include the following.

- Establishing the complexity of the lifting operation, including load and environmental complexity.
 - Load complexity will include characteristics such as weight, centre of gravity and presence of suitable lifting points. This can be established by an authentic and reliable source of information, measuring and weighing the load, or calculation.
Note: where the weight of the load cannot be accurately established, the notional weight must be multiplied by an appropriate factor (typically 1.5) to allow for possible inaccuracies.
 - Consideration of environmental hazards at the location of the operation will include the access and egress required for the lifting equipment, the location of overhead power lines or existing structures and the suitability of the ground to take the loads imposed by the lifting equipment during preparation for the lift and during the lift itself.
- Selecting the lifting equipment, based on the load characteristics (including weight of the load and any lifting points and accessories), the maximum height of lift and the maximum radius required. The rated capacity (safe working load (SWL)) of the lifting equipment should be specified by the manufacturer/installer in the user information supplied with the lifting equipment. **In cases where the lifting equipment has been derated at the time of thorough examination, the rated capacity (SWL) stated on the current report of thorough examination, issued by the competent person (equipment examination), should be used.** Manufacturer's sales leaflets and model classifications marked on the machine should not be relied on for the rated capacity (SWL) of specific lifting equipment.
- Ensuring that the lifting equipment is not operated in wind speeds in excess of those given in the instruction manual for the lifting equipment. The wind area of the load should also be taken into account to ensure that movement in the wind does not present a hazard.
- Ensuring that the lifting equipment has been thoroughly examined within the previous 12 months (or six months for equipment used for the lifting of persons). It is essential that the report of thorough examination, which confirms that the equipment is safe to use, is available. After installation or assembly ensure the equipment is inspected and checked before first use to confirm it has been installed correctly.
Note: the report of thorough examination for the lifting equipment should be carried with the machine.
- Selecting appropriate lifting accessories, including their method of attachment to the load, configuration and any protection used to prevent damage.

Environmental complexity (E)	3	Complex	Complex	Complex
	2	Intermediate	Intermediate	Complex
	1	Basic	Intermediate	Complex
		1	2	3
Load complexity (L)				

Relationship between complexity index and lift category

- Ensuring that lifting accessories have been thoroughly examined, within the previous six months, inspected and checked before use. It is essential that the report of thorough examination is available to confirm that the lifting accessories are safe to use.
- Ensuring that a system for reporting and rectifying defects is in place.
- Designating a suitable person to check the lifting accessories and any lifting points that are provided on the load to ensure they are free from any obvious defect before attaching the load to the lifting equipment.
- Ensuring that the outcomes of the planning process are recorded in a risk assessment and method statement, which should be signed by the competent person (lift planner).
Note: in many instances, a basic lift may be covered by a generic risk assessment and a generic method statement, provided that no additional hazards are identified on site.
- Selecting and defining the roles of the members of the lifting team. In many instances it may be possible to combine some of the roles of members of the lifting team.
- Briefing all persons involved in the lifting operation to ensure that the safe system of work described in the method statement is understood. All persons involved in the lifting operation should be instructed to seek advice from the competent person (lift planner) if any change is required to the lifting operation, or if any doubts about safety arise. If one or more handlines/taglines are required to give more control of the load, the competent person (lift planner) should designate persons to handle the lines.
- Where numerous repeat loads are to be lifted over a long period of time regular checking must be in place to confirm that no environmental changes have occurred requiring changes in the safe system of work.
- Ensuring that there is a lift (crane) supervisor designated to direct personnel and that the operation is carried out in accordance with the method statement.

The competent person (lift planner) and lift (crane) supervisor should be aware of the limits of their knowledge and experience concerning lifting operations and, when conditions exceed these limits, further advice should be sought.

7.4.4.2 Intermediate lift

For an intermediate lift, the duties of the competent person (lift planner) should include the following, in addition to the duties listed above.

- Identifying all significant hazards in the operating area, including any areas required for access or setting up of the lifting equipment.
Note: this may involve the competent person (lift planner) visiting site if there are any concerns about the detail and quality of the information with which the competent person (lift planner) has been provided.
- Ensuring that a site/task-specific risk assessment and method statement, detailing control measures for the identified risks, is prepared.
- Liaising with any other person or authority, as required, to overcome any hazard, by including any necessary corrective action or special measures in the safe system of work.
- Determining any requirement for extra personnel in addition to the lifting equipment operator (such as a slinger, signaller or dedicated lift (crane) supervisor).
- Considering the effect of the lifting operation on surrounding property or persons, including the general public, and arranging for appropriate action to minimise any adverse effects, giving appropriate notice to all persons concerned.

7.4.4.3 Complex lift

For a complex lift the duties of the competent person (lift planner) should include the following, in addition to the duties listed for basic and intermediate lifts.

- Identifying all exceptional hazards in the operating area, including any areas required for access or setting up of the lifting equipment. This will require the competent person (lift planner) to visit the location of the planned lifting operation as part of the planning process.
- Ensuring that the method statement includes the exact sequence of operations when lifting the load.
- Preparing a sufficiently detailed and dimensioned drawing of the site, lifting equipment and the load, identifying the load path, pick up and set down areas, together with the position of any exceptional hazards in the area (for example railway lines). The information provided should be sufficient to enable the operator to position the lifting equipment accurately.

It is good practice for the competent person (lift planner) to be present on the site during a complex lift.



Complexity of a lift

The case study in Appendix B illustrates the way in which both the complexity of the load being lifted and the environment in which the lift is taking place affect the overall complexity of the lift. The case study takes a typical construction lifting operation: the lifting of a generator and fuel tank.

Three different situations are evaluated, with examples of the hazards encountered and the control measures required to eliminate or reduce those hazards to an acceptable level.

It should be noted that this is an example, and does not identify all the hazards that may be present in a given circumstance.

LIFTING OPERATIONS

7.4.5 Carrying out a risk assessment

Having identified the hazards associated with the task, a risk assessment should be carried out to identify who might be harmed, the chance of them being harmed and the consequences of any harm.

This assessment should be recorded.

7.4.6 Identifying control measures

Once the risk assessment has highlighted the risks involved in the task, the procedures and measures required to control them should be identified and implemented.

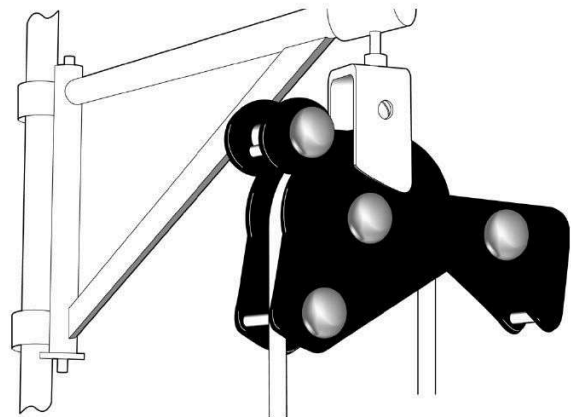
7.4.7 Developing the method to be used

Having identified the hazards, evaluated the risks and worked out the control measures required to carry out the task safely, these components should be developed into a coherent plan. Any contingency measures and rescue procedures should be included in the plan.

7.4.8 Selecting lifting equipment and lifting accessories



A proprietary inspection chamber lifting attachment



Inertia braked gin wheels are now common practice and far safer than traditional gin wheels

LOLER require that lifting equipment and lifting accessories are of adequate strength and stability for the load being lifted.

Consider the following when selecting lifting equipment and lifting accessories for a particular task.

- Weight, dimensions and characteristics of the loader (lifting equipment) and the loads to be lifted.
- Operation, speed, radius and height of lift and areas of movement.
- Number, frequency and types of lifting operation.
- Space available for lifting equipment access, deployment (including the space required for correct deployment of stabilisers), operation and storage for accessories and times of inactivity.
- Control position that will be most suitable for the lifting operation (the control position should be selected to ensure that the operator has an adequate view of the load path and is adequately protected from crushing hazards).
- Need for motion-limiting devices (such as slewing arc or height limiters).
- Effect of the operating environment on the lifting equipment.

7.4.9 Recording the method in a lift plan

Once the method has been developed, it should then be recorded in a lift plan. The length and detail of this document depends on the complexity of the task to be undertaken, and on the risks involved.

A simple, low-risk job (such as a routine delivery and unloading of bulk materials or blocks to site) might only require the use of a brief, generic plan, whilst a more complex and high-risk job (such as delivering, unloading, assembling and lifting into position a loading platform on a construction site) would require a more detailed job specific plan.

The method statement covering all planned lifting on a site is often referred to as the lift plan.



Telehandlers should have, and follow, a lift plan

The lift plan should include a lifting schedule, listing each type of item to be lifted, together with the following information.

- Item description.
- Weight.
- Dimensions.
- Lifting points and method of attachment.
- Type of lifting accessories to be used and configuration.
- Pick up and landing locations referenced to the site plan.

7.4.10 Communicating the plan to all persons involved

One of the most important aspects of successful planning is to ensure that the contents of the plan are communicated effectively to and between all parties involved, taking account of language differences. Arrangements should be made to ensure that copies of any method statements are given to the appropriate people and that others involved or affected by the operation are fully briefed. Similarly, any changes to the plan should be communicated to all parties.

7.4.11 Reviewing the plan before the job starts

Immediately before a job starts, the risk assessment and method statement (lift plan) should be reviewed to check if any aspects of the operation or working environment have changed and assess the effect that these changes could have on the safety of the people involved. Where changes are identified, the competent person (lift planner) should amend the lift plan and sign any significant changes with their initials. If any modifications to the plan are required these should be communicated to all those involved.

For further guidance on planning of lifting and installation refer to:

- **Code of Practice for safe use of cranes - Part 1 General and Part 3 Mobile cranes (BS 7121)**
- **HSE leaflet *Risk assessment. A brief guide to controlling risks in the workplace* (INDG163)**
- **CIRIA publication *Crane stability on site* (C703)**
- **Industry Lifting Lead Appointed Person Group (ILLAPG).**

7.4.12 Hired cranes

In construction, mobile cranes are generally hired by the user on site from the crane owner. The two ways of doing this are by hiring a crane (hired crane) or employing a contractor to carry out the lifting operation (contract lift). The differences are summarised below.

7.4.12.1 Hired crane (hired and managed)

With a hired lift you are hiring the crane and crane operator. You will be provided with a lift supervisor, slinger or an appointed person. The employing organisation should ensure the following.

- Carry out all work in accordance with BS 7121.
- Supply the competent person (lift planner).
- Plan the lift and operate a safe system of work.
- Confirm that the crane hired is of a suitable type and capacity.
- Check the credentials of the crane company and certification supplied.

The crane owner has a duty to provide a crane that is properly maintained, tested and certified, and a competent operator.

7.4.12.2 Contract lift (fully contracted)

With a contract lift you will not only be hiring the crane and operators, you will also be hiring the services of the crane company to organise and carry out the whole lift operation. The employing organisation should specify the following.

- That all work is to be undertaken in accordance with BS 7121.
- The ground bearing capacity.
- The dimensions and weight of the load to be lifted.
- That the lifting contractor is to supply the competent person (lift planner).
- What other information and/or services will be provided to the lifting contractor by the employing organisation.

The lifting contractor is responsible for the following.

- Supplying the competent person (lift planner).
- Planning the lift and operating a safe system of work.
- Organising and controlling the lifting operation.



If an individual or organisation does not have expertise in lifting operations they should not hire cranes but should opt for a contract lift.

Before entering into a contract, the employing organisation should satisfy itself that the contractor has the necessary competence to carry out the work.



For further detailed guidance on crane hire and contract lifts refer to the Construction Plant-hire Association's *Crane hire and contract lifting* (Best practice guide).

7.5 Siting of lifting equipment

For further detailed guidance on the siting of lifting equipment refer to:

- *Code of Practice for safe use of cranes – Parts 1, 3, 4 and 5 (BS 7121)*
- *Code of Practice for the safe use of construction hoists (BS 7212)*
- *Code of Practice for the installation, maintenance, thorough examination and safe use of mast climbing work platforms (MCWPs) (BS 7981)*
- *Code of Practice for the safe use of MEWPs (BS 8460)*
- *CIRIA publication Crane stability on site (C703)*
- *Strategic Forum Plant Safety Group publication Ground conditions for construction plant (2014).*

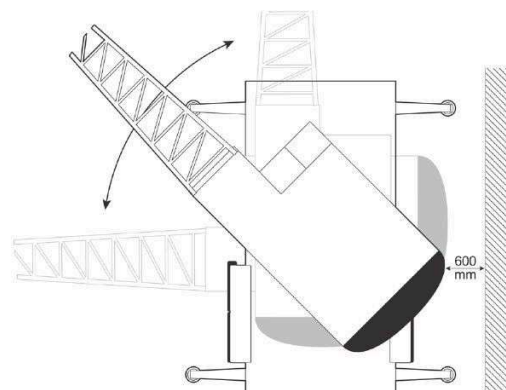
The area where lifting equipment is to be sited must be carefully assessed, before the equipment is brought to site and put into service, to ensure it is suitable. During this assessment the following points should be considered.

7.5.1 Clearances

The area chosen must be of a sufficient size to enable the lifting equipment to be manoeuvred into position, assembled, set up or installed, operated and stowed (fixed closed position), with sufficient clearances between the lifting equipment and surrounding structures, as detailed in the manufacturer's operation and instruction manual.

This is to ensure that crush zones are not created, and that damage does not occur to either the lifting equipment or the surrounding structures.

A gap into which persons may enter, which may be reduced by a slewing motion, should be at least 500 mm, but preferably never less than 600 mm (see diagram opposite).



Recommended clearance of 600 mm between moving parts and other structures

7.5.2 Ground conditions

! Insufficient consideration and assessment of ground conditions has been found to be a major cause of accidents with lifting equipment.

The stability of most mobile lifting equipment relies on the ground structure or surface on which it stands being able to withstand the loads and pressure imposed by the machine. Most lifting equipment manufacturers supply information on the loads imposed by the machine in the various operating and set-up configurations of the equipment.

These generally consist of the following.

- Dimensions of stabiliser support plate (pad).
- Ground-level pressure on stabiliser support plate.
- Maximum vertical load per stabiliser (or wheel for free-on-wheels machines).

! An assessment of the ability of the ground or surface to accept these loads should be made by a competent (appointed) person (lift planner).

This assessment may indicate that the proposed surface has insufficient bearing capacity to accept the loads imposed by the lifting equipment, in which case additional measures will need to be taken before the equipment can be set up. These may include using timber sleepers, proprietary mats or, in extreme cases, concrete pads to spread the applied load to an acceptable bearing pressure.

Where timber is to be used, it is important that the timber sections employed are of sufficient dimensions and strength to transmit the applied loads to the ground and that the timbers are pinned together to form a grillage (timber frame creating a foundation for the lifting equipment). The use of individual loose timbers has been found to be a major cause of accidents.

It is essential that any stabilisers are deployed in the manner specified by the lifting equipment manufacturer. When siting equipment, consideration should be given to the length of time that it will be erected in one position and the likely deterioration of supporting ground or foundation over time (such as timber rotting, undermining by water or frost, drying out and collapse of any adjacent excavations).

The hazards that need to be considered when assessing the ground or surface include the following.

- Underground services.
- Uncompacted fill.
- Paved areas.
- Open excavations.

- High water table.
- Basements.
- Cellars.
- Proximity to canals and rivers.
- Changes to site conditions during construction.
- Slope and camber of ground.

Details of any foundation or load spreading arrangements should be recorded in the method statement by the competent person (lift planner).



For further guidance on assessment of ground conditions and lifting equipment foundations refer to the CIRIA publications *Crane stability on site (C703)* and *Tower crane foundation and tie design (C761)* (CIRIA, 2018).

7.5.3 Overhead hazards

When siting lifting equipment care must be taken to ensure that the extending structure will not make contact with, or approach, overhead hazards (such as power lines, communications cables or overhead structures). Pre-planning of safe working procedures is important. The first step in avoiding danger is to find out whether there are any overhead power lines within or immediately adjoining the work area, or across any route or plane the crane may take.

If there are any overhead power lines over or near to the work area, the site boundaries, or over access roads to the work area, consult with the owners of the lines so that the proposed safe system of work can be discussed. Allow sufficient time for the line to be diverted or made dead, or for other precautions to be taken.



If any such lines are found it should be assumed that they are live until this has been proved otherwise by their owners.



For further information on working close to overhead power lines refer to Chapter D07 Underground and overhead services.

7.5.4 Proximity to railways and airports

If lifting equipment is to be sited adjacent to a railway or in the vicinity of an airfield or airport, the competent person (lift planner) should contact the railway or airport operator, as they may well impose restrictions on the height, lifting capacity and use of the lifting equipment.

The competent person (lift planner) should consult the aerodrome/airfield manager for permission to work if lifting equipment is to be used within 6 km of the aerodrome/airfield and its height exceeds 10 m over that of the surrounding structures or trees. When the project is near to a railway the competent person (lift planner) must contact Network Rail in all circumstances where the tower crane and its load can collapse within 4 m of a railway asset or property boundary.



The Air Navigation Order makes it an offence to act recklessly or negligently in a manner likely to endanger aircraft.

Further guidance on the use of lifting equipment in the vicinity of airfields or railways is given in the following publications:



- ***Operating tower cranes in the vicinity of aerodromes, notification and en-route obstacle lighting* (TIN039). Construction Plant-hire Association**
- ***Guidance to crane users on the crane notification process and obstacle lighting and marking.* Civil Aviation Authority**
- ***Requirements for tower cranes alongside railways controlled by Network Rail.* Construction Plant-hire Association**

7.5.5 Access to and egress from the site

It is important when siting lifting equipment to ensure that there is adequate access to the set-up position for both the machine and any supporting transport. It is equally important to ensure that adequate egress will be available when the lifting equipment is removed from site.



A self-erecting tower crane

A self-erecting tower crane was being used for lifting on a refurbishment project in a town centre. The crane stood in a courtyard for the duration of the project and the only way into the courtyard was through a single archway, which was just high enough to allow the folded self-erecting tower crane to pass below.

During construction a beam was inserted at the top of the arch to support a wall being constructed above. When it was time to remove the crane from site it was found that the headroom had been reduced to such an extent that the crane could not pass through the archway. The only way to remove the crane was to dismantle it at considerable cost and inconvenience to the site.

LIFTING OPERATIONS

7.5.6 Wind

The lifting equipment supplier or manufacturer will be able to advise on the maximum in-service and out-of-service wind conditions for the specific model to be used. It should be remembered that this is a design maximum wind speed and the operator may decide to take the equipment out of service at a lower speed due to the type of load being lifted or difficulty in controlling the equipment.

The operator has the primary responsibility for making the decision, in conjunction with the competent person (lift planner) or lift (crane) supervisor.



The operator's decision to take the equipment out of service should not be overridden by site management under any circumstances.

The wind forces exerted on both an item of lifting equipment and any load suspended from it may be quite large and affect both the strength and stability of the equipment, and safe handling of the load. It is not always appreciated that these forces are due to wind pressure, not wind speed, and consequently, if the wind speed doubles, the wind pressure increases by a factor of four. This means that a small increase in wind speed can have a significant effect on the safe operation of the lifting equipment.

Wind speeds should be monitored by either the anemometer installed on the equipment (such as tower cranes and large mobile cranes and mobile elevated working platforms (MEWPs) or the use of hand-held anemometers, bearing in mind that wind speed increases with height and that the wind pressure on the load may increase as it is lifted.

All operating personnel should be aware of the maximum in-service wind speed for the particular equipment being used. This value may need to be significantly reduced when lifting loads with a large wind sail area.

Tower cranes normally have a maximum in-service design wind speed of 45 mph (20 m/s, 72 kph), but following a review of in-service wind speeds by the CPA Tower Crane Interest Group, the industry-recommended maximum wind speed at which tower cranes operating in the UK must be taken out of service is 38 mph (16.5 m/s, 60 kph). This takes into account the time required to take the crane out of service and the difficulty of lifting large area loads in high winds. This maximum wind speed would be reduced by the competent person (lift planner), depending on the surface area of the load and other factors relating to the operation.

For mobile cranes, the duty charts will specify the maximum wind speed at which the mobile crane must be taken out of service. This value will vary with the specific make and model of crane and also the duty on which it is rigged, but normally has a maximum value of 31 mph (14 m/s, 50 kph) and is frequently below this value.



For further information on wind speeds and how they may be estimated by use of the Beaufort scale, refer to Appendix D.



For specific guidance on the effect of loads with large wind sail areas on limiting wind speeds, refer to the Liebherr publication *Influence of wind on crane operation*, which can be downloaded from its website free of charge.

7.5.7 Noise

Some locations where lifting equipment is used may have restrictions on noise, particularly at night, and the competent person (lift planner) should ensure that any such restrictions are taken into account in the planning process.

7.5.8 Exhaust gas emissions

If lifting equipment powered by an internal combustion engine is to be used in a confined space the competent (appointed) person (lift planner) should consider the effect of vehicle exhaust emissions on persons in the vicinity and take appropriate measures (such as forced ventilation or the use of fume extraction ducts).

7.6 Lifting persons

Some items of lifting equipment (such as MEWPs) or suspended cradles) are specifically designed to lift people.

The Approved Code of Practice to regulation 5 of LOLER says:



The raising and lowering of people using work equipment, which is not specifically designed for the lifting of people, should only be undertaken in exceptional circumstances, when it is not practicable to gain access by less hazardous means. Where it is necessary to use such work equipment then you should ensure that all necessary precautions are taken to ensure safety, including appropriate supervision.

If people are being lifted by lifting equipment not specifically designed for the purpose (in exceptional circumstances only), steps must be taken to ensure the following.

- The carrier is constructed to prevent a person using it from being crushed, trapped or falling from the carrier, either during lifting or carrying out work from the carrier.

- Personal fall protection equipment is worn.
- The equipment contains suitable devices to prevent the carrier falling.
- A person trapped in a carrier is not exposed to danger and can be rescued in the event of the lifting equipment breaking down.



For further guidance on MEWPs, refer to Chapter C08 Lifting equipment.



If a personnel carrying platform suspended from a crane is being used, BS 7121 Part 1 gives detailed advice.



MEWPs are specifically designed for lifting people

7.7 Control of lifting operations

When a lifting operation is undertaken it is essential that it is adequately controlled by suitably trained and competent people. Depending on the complexity of the lift the following roles may be involved.

Competent (appointed) person (lift planner). Although the competent (appointed) person (lift planner) has overall responsibility for the lift, they may choose to supervise the operation themselves or delegate the role to a lift (crane) supervisor.

Lift (crane) supervisor supervises the operation on behalf of the competent person (lift planner), working to the approved method statement or lift plan. If circumstances require any change to the method statement, this must be approved by the competent person (lift planner).

Operator operates the equipment within its permitted duties as per the method statement and as instructed by the lift (crane) supervisor.

Slinger/signaller connects lifting accessories to the load, initiates lifting by signalling to the operator, observes the path of the load to its destination, signals to the operator and detaches lifting accessories once the load is safely positioned.

7.7.1 Combination of roles

In certain circumstances it may be possible to combine some of the roles of members of the lifting team (illustrated in the following table). However, the table should not be taken as definitive for every circumstance. Role combination should only take place following review of the lifting operation by the competent person (lift planner).

It should be noted that a competent person (lift planner) is required to plan all lifting operations. It is good practice for them to be present on the site during a complex lift. In some circumstances it may be appropriate for the competent person (lift planner) to also assume other roles (such as lift (crane) supervisor, slinger/signaller or operator). The combination of roles requires that the person undertaking the combined role has achieved the necessary competence for each role.

Activity	Role	Lift category		
		Basic	Intermediate	Complex
Planning	Competent person (lift planner)	Required	Required	Required
Site visit	Competent person (lift planner)	Not essential	May be required	Required
Lifting operation	Competent person (lift planner)	Not essential	May be required	Required
	Lift (crane) supervisor	Roles may be combined	Roles may be combined	Required
	Operator			Required
	Slinger/signaller		Required	Required

Note: this table is for guidance. It is the competent person's (lift planner's) responsibility to determine the combination of roles for each lifting operation following a site-specific risk assessment that takes account of the load, the lifting equipment and the site environment.

7.7.2 Signalling methods

Communication between the people involved in the lifting operation should be clear and unambiguous. Communication can be by recognised hand signals (refer to Appendix C) or voice communication using portable radios. Signallers should be provided with distinctive personal protective equipment (such as fluorescent safety-helmet covers) so that they are immediately identifiable to the lifting equipment operator (Build UK sites have initiated the use of orange safety helmets for slinger/signallers and white for competent persons).



All CPA publications, including TIN 017 Radio communication for lifting operations, can be downloaded free of charge.

LIFTING OPERATIONS

7.7.3 Radio communication systems

Lifting equipment is often used on congested construction sites where the signaller is out of sight of the lifting equipment operator (blind lifts) and the standard hand signals cannot be used. As an alternative, hand-held very high frequency/ultra high frequency (VHF/UHF) radios are often used. This, however, can lead to a number of problems that may interfere with the clear communication vital for safe lifting operations.

Some examples are shown below.

- Loss of signal and thus communication, leading to loss of control of the lifting operation.
- Interference from radios on adjacent sites, which can lead to loss of communication to the lifting equipment operator.
- A misunderstanding between the lifting equipment operator and the signaller, leading to problems, such as a load being lifted before the slinger has their hands clear, loads colliding with a building and the load being lowered before people are clear of the landing area.

If communication is lost during a lifting operation, the operator must stop immediately until communication is regained.

The following points should be considered for the use of radio communication systems.

Radio specification: to avoid loss of signal and interference, select the correct radio equipment for the application.

Radio frequencies, channels, coverage, and license requirements: two types of radio system for two-way communication on construction sites are available. Low-output power private mobile network (PMR) radios can be used without a licence; higher output power radios require a licence from the Office of Communications (Ofcom).

Recording of voice communications: it is recommended that radio systems used on tower cranes are provided with systems that record all voice communications. The information may be stored on a local computer, server or remotely in the 'cloud'. The recording of voice communications assists with the monitoring and training of radio users, and the investigation of incidents. It is essential that crane operators and signallers are told that the radio system will record all voice communications, and the reasons why the recordings are being made.

Radio protocol, call signs and standard commands: poor radio discipline and failure to follow protocols can lead to lifting incidents. To avoid misunderstandings between the crane operator and signaller, all radio users should be trained and familiar with the radios and communication protocols that they will be required to use.

7.8 Operation of lifting equipment

Lifting equipment should only be operated by people who have been trained and assessed as competent.

7.8.1 Manuals and signs

All operations should be carried out in accordance with the manufacturer's operating manual, a copy of which should be with the lifting equipment at all times.

Checks should be made by the supplier to ensure that the manual is the correct manual for the lifting equipment supplied, conveys information to the users in a simple and understandable format and is in a language (usually English) that is understood by the operator.

All signs, labels and decals (stickers) on the lifting equipment must be clear, legible and in a language (usually English) that is readily understood by the operator. A rated capacity chart for the specific lifting equipment must be available to the operator.

7.8.2 Pedestrian-operated lifting equipment

The operation of some types of lifting equipment (such as lorry loaders and self-erecting tower cranes) is frequently carried out by a pedestrian operator at ground level, using remote controls.

Whilst pedestrian control provides flexibility, with the possible combination of roles there are several potential disadvantages to consider when planning the lifting operations. Operators may:

- be at risk of tripping and falling when trying to move around the site over uneven ground, whilst concentrating on controlling the lifting equipment or handling the load. Pedestrian-operated lifting equipment should only be controlled whilst the operator is stationary.
- not have a good view of the load and any obstructions and consequently must always have the lifting equipment jib or boom and load in sight at all times, unless they are working under the direction of a signaller who has a clear view of the load and load path
- put themselves in a hazardous position, where they may be struck by the load or part of the lifting equipment.



The operator should have full view of the jib and load at all times

Operating lifting equipment with remote controls

Operators should always:

- check site rules to confirm that the use of a remote control is permitted
- use the belt/neck strap provided
- be aware of the working area, including their proximity to the load and vehicle
- use the remote control in accordance with the manufacturer's instructions
- ensure the remote control is deactivated when not in use
- store the control unit in a suitable place when not in use.

Operators should never:

- stand between the load and a fixed object (such as a vehicle or a wall)
- walk whilst operating the remote control.

7.8.3 Radio remote controls

To prevent unauthorised use, the operator of lifting equipment that is controlled by transmitted signals (such as radio signals) should retain the control station (transmitter) in their physical possession or remove the key from its key-lock switch and, for short periods, retain the key in their possession. For longer periods, or when the lifting equipment is not in use, the transmitter should be kept in secure storage.

When the transmitter is fitted with a belt or harness, the operator should be wearing the harness before switching on the transmitter so that accidental operation of the lifting equipment is prevented. The transmitter should only be switched on when operating the lifting equipment and should be switched off before removing the harness and when attaching loads.

7.8.4 Rated capacity

The rated capacity (SWL) of lifting equipment should not be exceeded, except when testing the lifting equipment under the supervision of a competent, authorised person (usually after installation). Care should be taken to prevent pendulum swinging of the load, by controlling the operating motions to match the swing of the load and to keep it under control at all times.

Rated capacities apply only to freely suspended loads. The hoisting, slewing, telescoping, or raising and lowering motions of lifting equipment should not be used to drag a load under any circumstances. Before lifting a load, the hook and hoist rope should be vertically above the centre of gravity of the load. Failure to observe these points can adversely affect the stability of the lifting equipment or introduce loadings (stresses) into the lifting equipment structure for which it has not been designed and, even with a rated capacity indicator/limiter fitted, a sudden structural failure or overturn can occur without warning.

It is essential that all lifting equipment is clearly marked to indicate its rated capacity (SWL). Where the rated capacity (SWL) depends on the lifting equipment's configuration (such as changing radius) it should be clearly marked to indicate its rated capacity (SWL) for each configuration. As an alternative, information that clearly indicates its rated capacity (SWL) for each configuration should be kept with the lifting equipment.

Equipment designed for lifting persons should be clearly marked to confirm that it is suitable for lifting persons. Equipment which might be used in error for lifting persons must be clearly marked to indicate that it is not for lifting persons.

Most lifting equipment with a rated capacity not less than 1,000 kg or an overturning moment not less than 40,000 Nm is fitted with a load control device that will sense when the rated capacity is being approached or has been exceeded and prevent further movement. These devices vary between different types of lifting equipment and on cranes are generally known as rated capacity limiters (RCL). Cranes are also fitted with rated capacity indicators (RCIs), which indicate approach to overload and overload.

 **Effective lift planning should ensure that the rated capacity (SWL) of lifting equipment is never exceeded. The RCL or load control device should never be used to establish the weight of a load.**

7.8.5 Lifting of loads near persons

When loads have to be lifted in the vicinity of persons, extreme care should be exercised and adequate clearances allowed. The route of the load should be planned to prevent lifting over persons. Operators and signallers should pay particular attention to possible dangers of persons working out of sight.

All persons should be instructed to stand clear of the load being lifted. When lifting from a stack, they should be instructed to stand away from the stack in case adjacent materials or objects are displaced. This also applies to the removal of part loads from the deck of a vehicle.

Where possible, lifting of loads over pavements, highways, railways, rivers, or other places to which the public have access, should be avoided. If this is not possible, permission should be obtained from the appropriate authority and the area kept clear of traffic and persons.

LIFTING OPERATIONS

7.8.6 Multiple lifting

Multiple or tandem lifting (carrying out a lift with more than one piece of lifting equipment attached to the load) should always be planned and carried out with great care. A multiple lift is always classified as **complex**. Multiple lifts are generally only carried out with mobile cranes. Consequently, the term *crane* will be used here, rather than lifting equipment.

7.8.6.1 Weight of the load

The total weight of the load and its distribution should be either known or calculated. Where the information is taken from a drawing, allowance should be made for manufacturing tolerances.

7.8.6.2 Centre of gravity

Owing to the variable effect of manufacturing tolerances, variable density, and so on, the centre of gravity of the load might not be known accurately and the proportion of the load being carried by each machine could therefore be uncertain. If the centre of gravity is uncertain, the lift will be classed as a non-routine lifting operation and will require a more detailed lift plan.

7.8.6.3 Weight of the lifting accessories/attachments

The weight of the lifting accessories/attachments should be part of the calculated load on the cranes. When handling heavy or awkwardly shaped loads, the deduction from the rated capacity (SWL) of the lifting equipment to allow for the weight of the lifting accessories/attachments might be significant. The weight of the lifting accessories/attachments, and hook blocks, where appropriate, and its distribution should therefore be accurately known.

In cases where the hoist ropes are reeved around pulleys that are part of a specially designed lifting accessory/attachment (such as a lifting beam), the weight of the removed hook block and hook may be taken into consideration when determining the net weight of the lifting accessories/attachments.

7.8.6.4 Capacities of the lifting accessories/attachments

The distribution of the forces within the lifting accessories or attachments that arise during the lifting operation should be established. The lifting accessories/attachments used should, unless specially designed for the particular lifting operation, have a capacity margin well in excess of that needed for its proportioned load.



Special lifting accessories/attachments might be necessary to suit the maximum variation in distribution and direction of application of loads or forces that can occur during multiple lifting.

7.8.6.5 Synchronisation of lifting equipment motions

If the variations in the direction and magnitude of the forces acting on the crane during the multiple lift are to be kept to a minimum, it is essential that the crane motions are synchronised.

Lifting equipment/cranes of equal capacity and similar characteristics should therefore be used whenever possible. In practice, there is always some variation due to differences in response to the activation of the control system.

The rated capacity of a crane is calculated on the assumption that the load is raised and lowered in a vertical plane. The crane structure is designed to withstand any lateral loads imposed by accelerations in the various crane motions, but it is unsafe to rely on this lateral strength to withstand horizontal components of 'out-of-vertical' lifts.

If the cranes have dissimilar characteristics, it is unlikely that the motions will be accurately synchronised. Therefore, an assessment should be made of the effect of variation in verticality of the hoist ropes that could arise from inequalities of speed, together with a determination of the means for keeping such inequalities to a minimum.

7.8.6.6 Instrumentation

Instruments are available that constantly monitor the angle of the load, verticality and the force in any hoist rope throughout the lifting operation. The use of such instruments and the restriction of the motion speeds, together with the strict use of one motion at any one time, can assist in the control of the loads on the cranes within the planned values.

7.8.6.7 Supervision

One competent person (lift planner) should be in attendance and in overall control of a multiple crane operation. Only this person should give instructions to personnel operating or driving machines, except in an emergency when a commonly recognised stop signal may be given by any person observing a situation leading to danger.

If all the necessary points cannot be observed from one position, other personnel should be positioned at various points to observe and report to the person in charge of the operation.



It is essential that adequate means of signalling are provided between the person in charge of the operation, the operators of the cranes, and the slingers and signallers.

7.8.6.8 Recommended rated capacity during lifting

As all the factors cannot be accurately evaluated, an appropriate down-rating should be applied to all the cranes involved. The down-rating might need to be 20% or more.

7.9 Training and competence of personnel

As with all work activities on site, employers must ensure that the correct levels of training, competence and medical fitness of personnel are in place, and that maintenance and inspection regimes are carried out and recorded.



For detailed information, refer to Chapter C06 Mobile work equipment: 6.7 Training and competence of personnel, and 6.8 Maintenance, checks and inspections.



For further guidance on training and competence, refer to BS 7121 Parts 1, 3, 4 and 5.

For detailed guidance on medical fitness refer to the Strategic Forum for Construction Plant Safety Group's *Medical fitness to operate construction plant* (Good practice guide).

Appendix A - Definitions

Competent (appointed) person or lift planner. A person who has the competence, adequate training, authority and experience to take overall responsibility and control of a lifting operation, having been formally appointed in writing by the management of the organisation(s) that requires the load to be moved.

Competent person (equipment examination). (For the purposes of inspection and examination.) A person who has the practical and theoretical knowledge, together with actual experience of what they are to examine, so as to enable them to detect errors, defects, faults or weaknesses, which it is the purpose of the examination or inspection to discover; and to assess the importance of any such discovery.

Crane co-ordinator. A person who plans and directs the sequence of operations of cranes to ensure that they do not collide with other cranes, loads and other equipment (for example, concrete placing booms, telehandlers and piling rigs).

Lift (crane) supervisor. A person who controls the lifting operation and ensures that it is carried out in accordance with the appointed person's safe system of work (lift plan).

Crane operator. A person who operates the crane for the purpose of moving and positioning loads or erection of the crane.

Lifting accessory. A lifting beam or frame, chain sling, rope sling or similar gear, a ring, link, hook, interlocks, plate clamp, shackle, swivel or eyebolt, and any loose equipment that is used with lifting gear.

Lifting equipment. A piece of work equipment for lifting or lowering loads, including a crab, winch, pulley block or gin wheel (for raising or lowering), a hoist, crane, shearlegs, excavator, dragline, piling frame, aerial cable way, aerial ropeway or overhead runway, goods hoist, MEWP, scissor lift, vehicle hoist, ropes used for access, forklift truck, lorry loader and passenger lift.

Mobile crane. A crane capable of travelling under its own power, but does not include cranes that travel on a line of rails.

Plant and equipment. Any plant, equipment, gear, machinery, apparatus or appliance, or part thereof.

Rated capacity. The load that any item of lifting equipment (such as a crane or hoist) or any accessory for lifting is designed to lift for a given operating condition (such as configuration or position of load). Rated capacity was formerly known as safe working load (SWL).



On certain types of crane (for example, mobile cranes), the rated capacity includes the weight of the hook block and all the lifting gear. The weight of these should be deducted from the declared rated capacity of the lifting equipment to obtain the net load that can be safely lifted.

Rated capacity indicator/limiter RCI/L. Device(s) that warns of the approach to overload and prevents the crane from being overloaded as described in BS EN 12077-2.

Thorough examination. An examination by a competent person (equipment examination) in such depth and detail as the competent person considers necessary to enable them to determine whether the equipment being examined is safe to continue in use.